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Turbulent Heat Transfer over roughness: a comprehensive review of theories and turbulent flow structure

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ABSTRACT

Turbulent heat transfer over rough surfaces is crucial in natural and industrial processes, yet it has received less attention than that of momentum transfer. Recently, research in this field is rapidly growing, driven by advancements in computational methods, the role of surface roughness in meteorology, and the growing use of Additive Manufacturing technologies for energy and thermal management systems that inherently create rough surfaces. Despite these advances, a comprehensive understanding of turbulent heat transfer over rough surfaces, along with reliable predictive models applicable across various fields and roughness topologies remains lacking. The present paper reviews the key experimental and numerical studies to explore the mechanisms governing turbulent heat transfer over rough surfaces. It aims to guide future research and provide a solid foundation for understanding rough-wall turbulent heat transfer in various applications, including aerodynamics, meteorology, aerospace, and thermal management systems. By comparing with smooth-wall flows, the paper highlights how surface roughness affects turbulence structures and the thermal boundary layer. Existing models and scaling laws for heat transfer are critically evaluated, with attention to their applicability across different roughness types and flow conditions. The review also examines the analogy between heat and momentum transfer, the impact of roughness on thermal efficiency, and higher-order turbulent statistics. The literature shows that surface roughness breaks the Reynolds analogy, where any rise in Stanton number (*St*) is paired with a relatively larger increase in skin friction coefficient (C_f). As roughness Reynolds number (k_s^+) grows, the temperature shift ($\Delta\Theta^+$), which reflects heat transfer due to roughness, tends to flatten. This behaviour suggests a potential asymptotic thermally fully rough regime that is distinct from the velocity field's behaviour. However, the limited range of Reynolds (*Re*) and Prandtl (*Pr*) numbers achievable in numerical simulations allows only speculative conclusions about this asymptotic behaviour. Furthermore, the potential for a reduction in $\Delta\Theta^+$ at high k_s^+ values, possibly even becoming negative, remains feasible. The available heat transfer models do not account for either $\Delta\Theta^+$ trend, highlighting the need for a new theory or a revision of traditional models. It is hypothesised that roughness enhances heat transfer by channelling energy toward colder, high-velocity regions, indicating the importance of roughness thermal conductivity. It is suggested that a universal rough-wall heat transfer theory should incorporate factors including roughness topology, roughness thermal conductivity, k_s^+, and *Pr*. In addition to expanding the range of *Re* and *Pr*, a systematic examination of various roughness types and topologies is necessary, as the temperature remains sensitive to specific geometric details.

List of Abbreviations: ABL, Atmospheric boundary layer; CFD, Computational fluid dynamics; CFS, Coherent flow structures; CSL, Conduction sublayer; CV, Cavity vortex; DA, Double average; DNS, Direct numerical simulation; erf, Error function; ES, Effective slope; IAEA, International atomic energy agency; KB, Kolmogorov–Brutsaert; KH, Kelvin-Helmoltz; LES, Large eddy simulation; PDF, Probability density function; RA, Reynolds analogy; RANS, Reynolds-averaged Navier-Stokes; RMS, Root mean square; SBL, Stable boundary layer; TKE, Turbulent kinetic energy; TPMS, Triply periodic minimal surfaces; VSL, Viscous sublayer.

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Nomenclature

Symbol	*Description SI unit*
A_t	total plane area (wall-projected area or area of the smooth surface) m^2
A_p	Projected plane surface area m^2
A_f	Projected frontal area m^2
A_w	Total wetted surface area m^2
A_s	Sheltered surface area m^2
A_e	Exposed surface area m^2
A_φ	Fluid-occupied area m^2
B	Roughness constant -
C	Smooth wall intercept -
C_f	Skin friction coefficient, $C_f = \tau_w / \left(\frac{1}{2}\rho U_{ref}{}^2\right)$ -
$C_{f,S}$	Skin friction coefficient for smooth surfaces -
$C_{f,R}$	Skin friction coefficient for rough surfaces -
c_p	Specific heat capacity J/kgK
$\mathscr{D}$	Mass diffusivity m^2/s
d	Zero-plane displacement height or wall offset m
ES	Effective Slope of roughness -
G	Mass concentration kg/m^2
g_k	Roughness similarity function -
h	The height of regular roughness element, such as riblets m
h_c	Centred height-map m
TKE	Turbulent kinetic energy m^2/s^2
k	Roughness height (regular roughness), roughness height map (irregular roughness) m
k_s	Equivalent sand-grain roughness height m
$\overline{k}$	Mean height of the roughness m
k_s^+	Roughness Reynolds number, $k_s^+ = h_s u_\tau/\nu$, -
k_t^+	Roughness Reynolds number, $k_t^+ = S_t u_\tau/\nu$, -
Nu	Average Nusselt number, $Nu = hD_h/\lambda_f$ -
P	Probability density function
p	riblet spacing (pitch) m
p^+	Friction-scale dimensionless longitudinal riblet spacing -
Δp	Pressure drop/ pressure loss Pa
Pr	Molecular Prandtl number, $Pr = \nu/\alpha$ -
Pr_t	Turbulent Prandtl number -
$Pr_{t,eff}$	Effective turbulent Prandtl number -
q	Heat flux W/m^2
q_w	Wall heat flux W/m^2
RA	Reynolds Analogy factor -
RA_s	Reynolds Analogy factor for smooth surfaces -
RA_R	Reynolds Analogy factor for rough surfaces -
Re	Reynolds number, -
Re_b	Bulk flow Reynolds number, $Re_b = U_b\delta/\nu$ -
Re_τ	Friction Reynolds number, $Re_\tau = u_\tau\delta/\nu$ -
S_a	Arithmetic-averaged height of roughness m
S_q	Root-mean-square height of roughness m
S_{sk}	Skewness of roughness -
S_{ku}	Kurtosis of roughness -
S_z	N-point average peak-to-valley height m
S_v	Minimum roughness height (Depth of the deepest valley) m
S_p	Maximum roughness height (the height of the highest peak) m
S_t	Maximum peak-to-valley height, $K_t = K_v + K_p$ m
St	Stanton number, $St = NuRe^{-1}Pr^{-1} = q_w/(\rho c_p U_{ref}\Theta_{ref})$ -
St_k	Roughness Stanton number -
St_c	Cavity Stanton number -
$St_{k,s}$	Sheltered roughness Stanton number -
$St_{k,e}$	Exposed roughness Stanton number -
Sc	Schmidt number -
Sh	Sherwood number -
S_{ij}	Stress tensor Pa
T_{ref}	Reference temperature K
T	Mean Temperature K
T_w	Wall temperature K
T_K	Temperature at the outer edge of roughness sublayer K
T_a	Fluid arithmetic mean temperature K
T_b	Fluid bulk temperature K
T_m	Mixed-mean temperature K
T_∞	Free stream temperature K
T'	Instantaneous (fluctuating) temperature K
t_η	Kolmogorov time scale s
u	Streamwise velocity m/s
u'	Streamwise instantaneous (fluctuating) velocity m/s
u_g	Velocity at roughness peaks m/s
u_j	Velocity component, $u_1 = u$, $u_2 = v$, $u_3 = w$ m/s
$\overline{u}_j$	Reynold-average velocity component m/s
u_i'	Instantaneous velocity component m/s
U	Mean velocity m/s
U_{ref}	Reference velocity m/s
U_∞	Free-stream velocity m/s
U_b	Bulk flow velocity m/s
U_c	Centreline velocity m/s
u_τ	Friction velocity, $u_\tau = \sqrt{\tau_w/\rho}$ m/s
U_{ref}^+	Friction-scale dimensionless reference velocity -
U^+	Friction-scale dimensionless velocity, $U^+ = U/u_\tau$ -
U_e^+	Friction-scale dimensionless velocity in the outer layer -
U_R^+	Friction-scale dimensionless velocity for rough surfaces -
U_S^+	Friction-scale dimensionless velocity for smooth surfaces -
ΔU^+	Roughness function -
$\left\langle u_i'^+ u_j'^+ \right\rangle$	Normalized Reynolds stress tensor -
$\left\langle u_i'^+ \theta'^+ \right\rangle$	Normalized heat flux component -
v	Normal wall velocity m/s
v'	Normal wall instantaneous (fluctuating) velocity m/s
y	Wall normal distance m
y_0	Momentum roughness length m
y_e	Modified wall-normal distance m
$y_{0,\theta}$	Thermal roughness length scale m
y^+	Friction-scale dimensionless wall distance, $y^+ = y\, u_\tau/\nu$ -
w	Normal-wall velocity m/s
w'	Spanwise instantaneous (fluctuating) velocity m/s
h_i	Heat transfer coefficient W/m^2K
$\overline{h}$	Average heat transfer coefficient W/m^2K

Greek letters

α	Thermal diffusivity, $\alpha = \lambda/\rho c_p$ m^2/s
α_h	Local roughness angle, rad
α_c	Roughness constant -
$\beta(Pr)$	Prandtl number-dependent offset function -
δ	Momentum boundary layer thickness m
δ_e	Effective momentum boundary layer thickness m
δ_t	Thermal boundary layer thickness m
δ_t^+	Friction-scale dimensionless thermal boundary layer thickness -
δ_θ^+	Friction-scale dimensionless conduction sublayer thickness -
δ_{ij}	Kronecker Delta -
ε	Turbulent dissipation rate m^2/s^3
η	Kolmogorov length scale m
Θ_{ref}	Temperature difference, $\Theta_{ref} = T_{ref} - T_w$ K
Θ	Mean Temperature, $\Theta = T - T_w$ K
Θ_τ	Friction temperature, $\Theta_\tau = q_w/\rho c_p U_\tau$ K
Θ^+	Friction-scale dimensionless temperature, $\Theta^+ = \theta/\theta_\tau$ -
Θ_k	Temperature at the level of roughness K

Θ_θ	Conduction sublayer temperature K
Θ_θ^+	Friction-scale dimensionless conduction sublayer temperature -
Θ_m	mixed-mean temperature K
Θ_R^+	Friction-scale dimensionless temperature for rough surface -
Θ_S^+	Friction-scale dimensionless temperature for smooth surface -
$\Theta_{a,R}^+$	Outer layer friction-scale dimensionless temperature for rough surface -
$\Theta_{a,S}^+$	Outer layer friction-scale dimensionless temperature for smooth surface -
λ_f	Frontal solidity parameter of roughness $J/kg.K$
λ_p	Plan solidity parameter of roughness $J/kg.K$
Λ	Roughness density parameter -
μ	Dynamic viscosity $kg/m.s$
μ_t	Turbulent viscosity $kg/m.s$
ρ	Density kg/m^3
ν	Kinematic viscosity m^2/s
τ	Dimensionless time scale, t/t_η -
τ_w	Wall shear stress Pa
φ	Void fraction -
Φ	A typical flow filed variable
h_i	Heat transfer coefficient W/m^2K
$\overline{h}$	Average heat transfer coefficient W/m^2K

Special characters

κ	von Karman constant -
κ_θ	von Karman constant for temperature -
ε_H	Thermal eddy diffusivity m^2/s
ε_M	Momentum eddy diffusivity m^2/s
$\langle\dot{-}\rangle$	Double Average (intrinsic averages) -
$(\dot{-})$	Temporal average -

1. Introduction

Most convective heat transfers in both natural and industrial settings involve turbulent flows, where momentum, energy, and other properties are transported by turbulent energy transfer from large-scale to small-scale eddies through a process known as energy-cascading [1]. This process contributes to the chaotic and unpredictable behaviour of turbulent flows [2]. Despite extensive research, our understanding of the physics of turbulent flow, particularly in the presence of rough walls, remains incomplete. Current knowledge is primarily based on statistical and phenomenological theories [3,4], supported by experimental data and high-fidelity numerical simulations. These theories have been employed by various researchers to predict friction drag and heat transfer in wall-bounded turbulent flows.

Rough-wall turbulent heat transfer is a common phenomenon across a diverse array of natural processes and engineering applications, where surface roughness can significantly influence costs, emissions, and energy efficiency. This includes roughness from erosion and deposition on gas turbine components and internal combustion engines [5–11], heat transfer during ice accretion on wind turbine blades [12–16] and aircrafts [17–20], roughness on heat exchangers/sinks or thermal management devices fabricated by additive manufacturing [21–26], topological features and vegetations on the Earth's surface affecting heat exchange between the Earth's surface and the atmosphere in meteorology and the Atmospheric Boundary Layer (ABL), and complex terrains in renewable energy [27]. Additionally, enhancing heat transfer, such as improving cooling efficiency, is crucial for the performance of numerous industrial devices [28–30]. Roughness can influence the wettability of surfaces in both natural and engineering systems, leading to complex flow dynamics [31], a subject that remains poorly understood [32]. In microfluidics for thermal management, the infinitesimal length scale of the system makes roughness a critical factor influencing performance [33]. Artificially generated surface roughness, characterised by distinct topological features such as ribs, grooves, pins, and fins, has been widely utilised for passive heat transfer enhancement [34–39]. Despite extensive research, the effectiveness of roughness in achieving substantial heat transfer improvement has yet to be conclusively proven.

In practice, roughness involves topographical features ranging from a few micrometres (e.g., from 5 μm up to nearly 40 μm on the surface of metallic additive-manufactured components [22]) to a few millimetres (e.g., approximately 1 mm on gas turbine blades or greater for biofouling on ship hulls) and can be as large as vegetation and tall buildings and mountain ranges in meteorology and the ABL. As schematically shown in Fig. 1, roughness can be classified into 2D and 3D roughness [40,41]. The term 2D roughness refers to surfaces with a linear profile, varying in only one direction, such as spanwise or streamwise ribs and grooves with different cross-sections. This type of

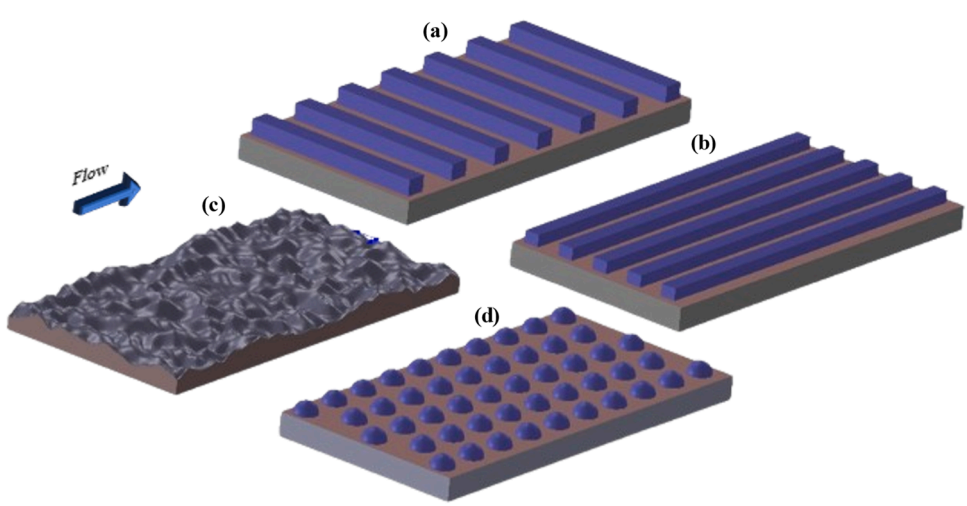

Fig. 1. Examples of 2D and 3D roughness elements are (a) 2D roughness with spanwise riblets, (b) 2D roughness with streamwise riblets, (c) 3D irregular roughness, and (d) 3D regular roughness with hemispheres.

roughness is artificially designed for heat transfer enhancement in various applications [42]. The term 3D roughness refers to surfaces with areal height maps consisting of regular or irregular roughness elements that are systematically or randomly distributed. Irregular 3D roughness can occur naturally as a consequence of manufacturing processes [43] or mechanisms such as erosion, corrosion, and deposition, as seen in energy systems [5], gas turbines [44], ice accretion on aircraft wings [45], deposit in combustion chambers [46], and biofouling on ship hulls and marine machinery [47]. Both 2D and 3D roughness have been widely employed for improving aerodynamic performance [48] and heat transfer enhancement [49].

Despite significant advancement, rough-wall turbulent flow and heat transfer remain incompletely understood, as highlighted by comprehensive literature reviews [50–52]. The challenge of rough-wall convective heat transfer involves predicting both flow friction (drag) and heat transfer. Currently, no universal predictive method exists due to insufficient understanding and the varying effects of different roughness topologies on flow. The uncertainties arising from this lack of reliable prediction incur costs amounting to billions of dollars annually [51,53]. Consequently, roughness continues to be a dynamic and prolific field of research.

While rough-wall momentum transfer has been studied for over a century [52], the first documented research of rough-wall heat transfer was conducted by Cope [54] in 1941, focusing on 2D roughness elements using air. The 1960s saw the development of some theories of rough-wall heat transfer through experimental work, while recent works have benefitted high fidelity simulations as well [4,55–63]. Despite the long history, research findings on heat transfer have lagged significantly behind those on momentum transfer (friction drag), as illustrated in Fig. 2. Chung et al. [51] highlighted that heat transfer remains one of the primary challenges in the study of rough-wall turbulent flows.

As evidenced in Fig. 2, research on rough-wall heat transfer has recently become more active, which can be attributed to three major factors. Firstly, the advancement of simulation techniques and the significant rise in computational power have made scale-resolving computational fluid dynamics (CFD) [46,55,60–68], including Direct Numerical Simulation (DNS) and Large Eddy Simulations (LES), more accessible. The numerical methods can visualise the flow dynamics by providing the distribution of the field's variables, such as velocity and temperature, that is impossible or difficult to achieve with current experimental methods. They also provide high degree of flexibility to systematically study rough-wall flows by altering roughness topology and flow parameters. Secondly, ever-increasing progress in additive manufacturing (3D printing) technologies has created new opportunities for heat transfer applications, such as in gas turbine components [22,26,69–77], electronic cooling [78–83], and heat sink and heat exchangers

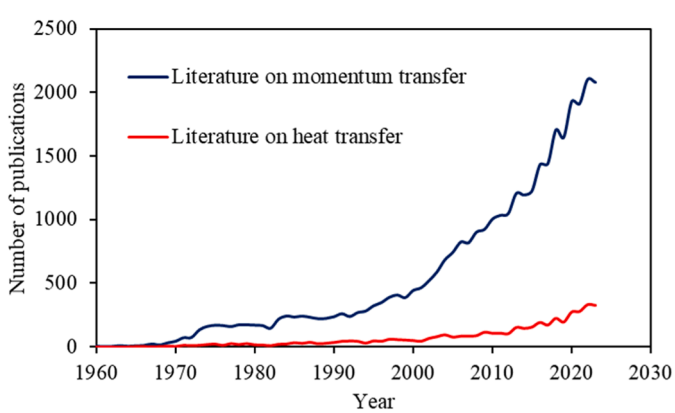

Fig. 2. Comparing the number of publications per year on rough-wall heat and momentum transfer. The source of data is *scopus.com* and data were obtained by separate searching of keywords "heat + flow + roughness" for heat transfer and "flow + roughness" for momentum transfer.

[82,84–89]. Therefore, considering additive manufacturing roughness, which is due to natural consequence of the process, has become of great importance [90–94]. Lastly, growing global concerns about climate change have increased the need for improved heat transfer solutions in renewable energy [37,95–101] and energy efficiency [102], as well as for technologies aimed at reducing the global carbon footprint [103–106]. This has created opportunities for implementing passive heat transfer techniques, such as artificial roughness, and complex designs, including lattice structures, intricate channels, and triply periodic minimal surfaces (TPMS), fabricated through additive manufacturing, which inherently produces natural roughness on the surfaces of these structures.

Accurate and reliable correlations for both friction factors and heat transfer coefficients are crucial for engineering calculations. Developing these correlations requires high-quality experimental data, high-fidelity numerical simulations, and in-depth studies to enhance the understanding of flow dynamics. Empirical correlations and models may be integrated into numerical simulations for practical engineering purposes (e.g., the models developed by Aupoix [107–109], Radenac et al. [110], Aupoix & Spalart [111], and Lee [112]). Developing models requires comprehensive understanding of the structure of turbulent flow, as illustrated in Fig. 3, which is the focus of the present research.

Several textbooks offer information on flow and heat transfer over surface roughness, including those authored by Schlichting & Gersten [113], Wilcox [114], Lin [115], Kays, Crawford & Weigand [116], and Saha et al. [40]. Previous reviews [49–52,117–119] have mainly concentrated on the effects of roughness on flow dynamics. Jiménez [50] reviewed experimental studies on rough-wall turbulent flows, introducing the d-type and k-type roughness classifications. Flack and Schult [117] examined predictive methods and correlations for frictional drag, while Flack and Schultz [120] explored outer-layer similarity. Raupach and Thom [121], Nepf [122], and Finnigan [123] reviewed studies on ABL flows, with a focus on dense roughness. García-Mayoral et al. [118] reviewed engineered surfaces designed for drag reduction. Piomelli [119] assessed numerical studies conducted at Queen's University's Turbulence Simulation and Modelling Laboratory in Canada. Chung et al. [51] provided an overview of advances in rough-wall turbulent flows, and Kadivar et al. [52] reviewed the fundamentals and theories underlying these phenomena.

Although recent reviews [51,52] have touched on heat transfer to some extent, they have primarily concentrated on momentum transfer and flow drag due to roughness. The first literature survey on the Nusselt number and pressure drop over rough surfaces was conducted by Bhattacharyya [124] in 1964, published by the International Atomic Energy Agency (IAEA). Subsequently, Bons [6,125,126] reviewed heat transfer and flow friction in gas turbine components, roughened due to degradation. Ligrani [127] provided an overview of various heat transfer augmentation technologies, including roughness, for cooling gas turbine components. Other researchers have focused on passive heat transfer enhancement using artificial 2D and 3D roughness elements [34–39,128]. Soleimani and Eckels [49] explored drag reduction and heat transfer enhancement through riblet roughness in channel flows. Rowin et al. [129] reviewed measurement equipment and methods for accurately predicting heat transfer over rough walls. Hantsis & Piomelli [57] examined recent high-fidelity numerical studies of passive scalar transport over rough walls, with a focus on turbulent flow properties and budgets.

To the best of the author's knowledge, no comprehensive review currently exists that thoroughly addresses the theory and fundamental analysis of rough-wall turbulent heat transfer. This study aims to fill that significant gap by offering a critical review of the experimental, numerical, and theoretical studies on turbulent convective heat transfer over rough surfaces. The review provides an overview of both historical and recent research in this area, examining key theories, the structural aspects (as shown in Fig. 3), and the effectiveness of surface roughness in enhancing heat transfer. This review paper aims to address critical gaps

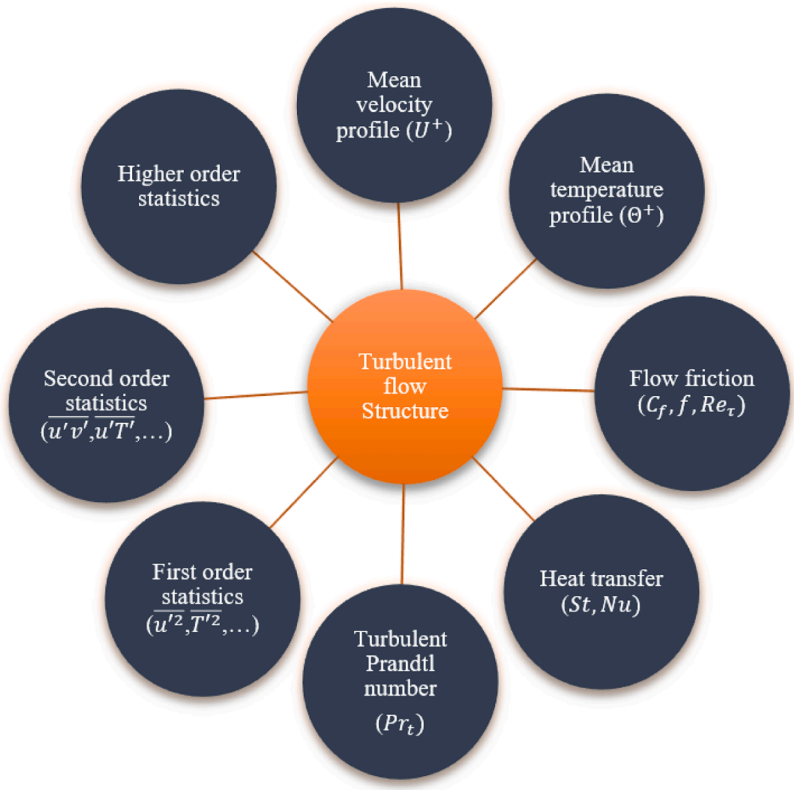

Fig. 3. Turbulent flow structure

in the understanding of turbulent heat transfer over rough surfaces, which has been investigated in scattered studies but lacks a unified, systematic analysis. Despite a number of studies on rough-wall heat transfer, the literature remains limited in exploration of roughness topology, Prandtl number, and thermal conduction characteristics, as well as the practical implications for industrial applications. This review provides a much-needed consolidation of existing findings, examining strengths, limitations, and unresolved questions. Specifically, the novelty of this paper is demonstrated through the following objectives:

- It provides a detailed overview of the state-of-the-art on rough-wall heat transfer and examine the analogy between heat and momentum transfer in turbulent flows, identifying where the current understanding is robust or needs refinement.
- This review examines the impact of roughness topological parameters, Prandtl number, and Reynolds number, and how these factors interplay to influence the thermal boundary layer and roughness efficiency.
- It critically evaluates various scaling laws and empirical correlations in light of the proposed theories and hypotheses, comparing them with observed data to highlight discrepancies, limitations, and areas that demand further investigation.
- Finally it draws informed conclusions, outlines current limitations, and proposes directions for future research.

2. Rough-wall convective flow

In the literature, heat and mass transfer are often treated as analogous scalar quantities, enabling findings from one field to be applicable to the other. For instance, temperature (T) findings can be transferred to mass concentration (c), and heat flux (q) findings to mass flux (G) by substituting molecular diffusivity (α) with mass diffusivity ($\mathscr{D}$) and setting the heat capacity (c_p) to one. Dimensionless numbers can also be used interchangeably; for example, the Nusselt number (Nu) can be used in place of the Schmidt number (Sc), and the Stanton number (St) can be substituted for the Sherwood number (Sh). Mass transfer is particularly crucial in ABL flows, where interactions with surfaces such as water, land, vegetation, sand, and urban areas involve the transfer of momentum, heat, and mass. This area of study, known as interfacial transport phenomena, plays a significant role in the global balance of gases such as carbon dioxide, methane, nitrous oxides, and dimethyl sulfide [130–145]. Given that much of the existing literature emphasizes heat transfer, this study also prioritizes heat transfer analysis to build on and contribute to the established body of knowledge. Nonetheless, the principles and findings are equally relevant to scalar transport such as mass transfer.

2.1. Hydraulically rough flow

A boundary layer flow is classified as hydraulically rough when the surface roughness elements are large enough to disrupt and influence the smallest turbulent eddies near the wall. According to phenomenological theory [146,147], these smallest eddies are characterised by the Kolmogorov length scale, $\eta = (\nu^3/\varepsilon)^{1/4}$, and the time scale, $t_\eta = (\nu/\varepsilon)^{1/2}$, where ν is the kinematic viscosity and ε is the turbulent dissipation rate. The relative size of the roughness length scale (k) compared to η determines the extent to which roughness impacts the flow. If $k < \eta$, the roughness is confined within the viscous sublayer (VSL), where viscous forces dominate and effectively dampen any disturbances caused by the roughness, resulting in a hydraulically smooth flow. Conversely, when $k > \eta$, the roughness disrupts the energy cascade, generating roughness-scale eddies that enhance turbulent mixing and increase both viscous and form drag.

Pioneering studies by Hagen [148] and Darcy [149] established the

influence of surface roughness on pressure drop, which they attributed to increased drag force and blockage effects [150]. In the 1930s, Nikuradse's experiments [151] laid the foundation for roughness models by providing detailed measurements of friction factors and velocity profiles in pipes with sand-grain roughness. Building on this concept, Colebrook and White [152,153] developed the widely used Colebrook-White equation to calculate the Darcy friction factor in rough pipes. In 1937, Schlichting further advanced the field by introducing the concept of equivalent sand-grain roughness (k_s) to quantify the effect of surface roughness on turbulent flows. Since then, k_s has been widely utilised as a rough flow length scale in related studies. More information can be found in Kadivar et al. [52] and Chung et al. [51].

A key characteristic of hydraulically rough flow is the partial or complete destruction of the viscous sublayer, which is replaced by a roughness sublayer (see Fig. 4). In the roughness sublayer, enhanced turbulent mixing results in a more uniform distribution of flow-averaged properties, such as velocity and temperature, enhancing the transport of momentum (increased drag), heat, and mass. Jiménez [50] and Raupach et al. [154] suggested that the roughness sublayer typically extends from the wall to approximately $2k - 5k$ [154], while Flack et al. [155] proposed that it spans from $3k_s - 5k_s$ for the roughness sublayer. It is worth mentioning that k refers to the roughness length scale, representing the roughness height, while k_s denotes the rough-flow length scale, representing both the roughness topology and the flow properties. The advantage of using k_s lies in its ability to incorporate flow characteristics when defining the roughness sublayer; however, it cannot be determined solely based on the roughness topography.

2.2. Problem definition

The rough-wall turbulent convective heat transfer is summarised in Fig. 4. The primary quantities of interest are the wall shear stress, τ_w, and wall heat flux, q_w. The former represents the drag force per unit area, while the latter denotes the heat transfer rate per unit area. Flow at wall distance y is characterised by the mean velocity profile $U(y)$ and mean temperature profile $\Theta(y)$, with flow properties including ν, α, ρ, and c_p, which represent kinematic viscosity, thermal diffusivity, density, and specific heat capacity, respectively. The temperature profile is given by $\Theta(y) = T(y) - T_w$, where $T(y)$ is the mean temperature at the wall distance y and T_w represents the wall temperature.

Let δ be the flow length scale, which is the characteristic length of the flow, which can be defined as the half height of the channel in channel flows, the radius of the channel in pipe flows, and momentum boundary layer thickness in boundary layer flows. The friction velocity is defined as [2]

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}}, \tag{1}$$

and subsequently, the friction temperature can be defined as [2]

$$\Theta_\tau = \frac{q_w}{\rho c_p u_\tau} \tag{2}$$

Asymptotic matching analysis suggests that the boundary layer, whether over smooth or rough walls, consists of two overlapping regions: an outer layer that scales with u_τ and δ, and an inner layer that scales with u_τ and a set of length scales characterising the surface roughness. Therefore, the friction (viscous) length scale ν/u_τ, can be employed to define dimensionless variables in the boundary layer. The friction-scale mean velocity, temperature, and wall distance are defined as $U^+ = U/u_\tau$, $\Theta^+ = \Theta/\Theta_\tau$, and $y^+ = yu_\tau/\nu$, respectively. Assuming the flow is in an "equilibrium" state, where δ and u_τ vary slowly in the streamwise direction, the problem can be expressed in dimensionless friction scales as finding $\tau_w/\left(\rho U_{ref}^2/2\right)$ and $q_w/\left(\rho c_p U_{ref}^2\,\Theta_{ref}\right)$, representing the dimensionless flow friction and heat transfer, respectively [51]. U_{ref} and Θ_{ref} are the reference velocity and temperature, respectively, which depend on the flow type. Roughness can be represented by using a height map, $k(x, z)$, where x, and z are the streamwise and the spanwise coordinates, respectively. This height map denote the distribution of roughness height, from which various roughness parameters can be derived.

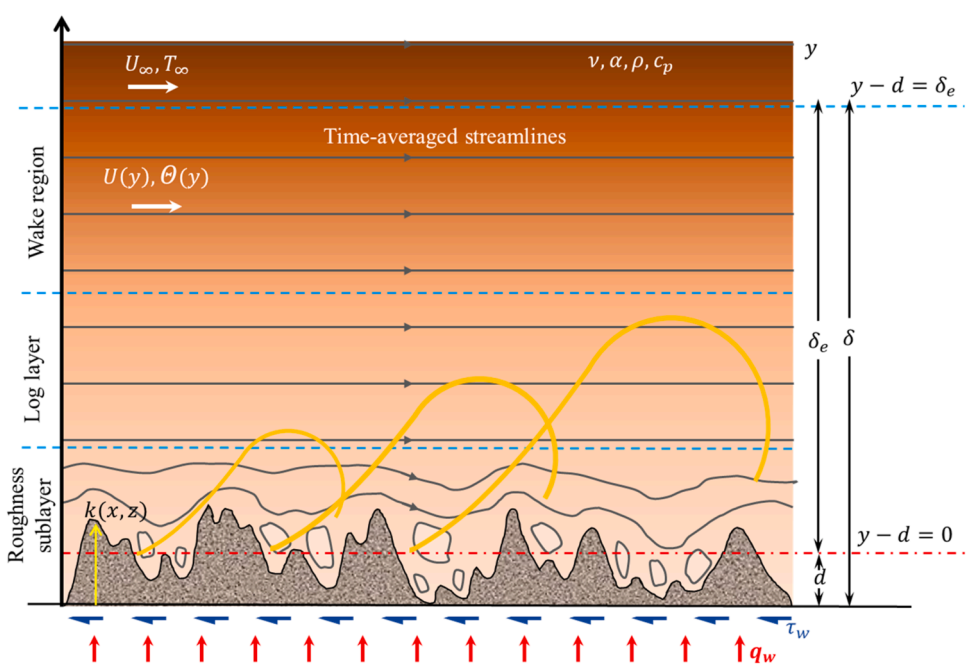

Fig. 4. Schematic of rough-wall boundary layer problem. Here, τ_w and q_w represent the wall shear stress and wall heat flux, respectively, for a given mean velocity profile $U(y)$ and mean temperature profile $\Theta(y)$ at wall distance y with flow properties ν, α, ρ, and c_p. The height-map of roughness is $k(x, z)$ and the wall offset is located at $y = d$.

2.3. Wall origin and virtual wall

Since roughness moves the entire flow upward [51,52], the choice of the origin of the wall-normal coordinate is crucial. The wall origin is defined as the level obtained by smoothing out the roughness geometry to create an idealised smooth surface [156]. This new reference level on the virtual wall is referred to as the wall origin and is denoted by d, also known as the wall offset. As a result, the outer turbulent flow perceives its origin at $y = d$ rather than $y = 0$ (see Fig. 4).

The wall offset is an important factor in wall similarity profiles (discussed further in section 9). To resolve the flow-physical inconsistency introduced by roughness, the shifted wall-normal coordinate $y - d$ is used. This adjustment ensures that all profiles in the outer layer properly overlap, adhering to the concept of wall similarity. The selection of the zero-plane displacement is further detailed in Chung et al. [51] and Chan-Braun et al. [157]. While the velocity at $y = 0$ is zero due to the no-slip boundary condition, the velocity at $y = d$ better reflects the actual flow dynamics in the presence of roughness.

The wall origin becomes particularly crucial when the roughness represents a significant portion of the boundary layer thickness. Thom [158] and Jackson [156] proposed positioning the wall origin at the centre of the drag profile on the roughness element. This approach involves adjusting the rough-wall logarithmic velocity profiles to eliminate negative velocities [159]. This adjustment is known as the "zero-plane displacement height," an important concept in meteorology and ABL flows [3]. Analogous to Jackson's drag-centroid (d), the centre of the heat source distribution from the roughness elements, denoted by d_θ, can be used for the origin of the temperature profiles [3]. However, determining d_θ can be challenging, often resulting in different wall origins for the velocity and temperature fields. As a result, most studies recommend using the velocity wall origin as the reference for the temperature wall origin (i.e. $d_\theta = d$).

The drag-centroid approach has been criticized for being potentially inappropriate in certain contexts [51,62,157,160–163]. Another widely used method for determining the wall origin involves fitting the law of the wall to the logarithmic velocity profile (see Section 4) [154,164–166]. This approach is straightforward and effective for high Reynolds number flows, adhering to wall similarity principles. However, for lower Reynolds numbers, the applicability of the logarithmic profile may be limited.

An alternative method for determining the wall origin involves using the mean height of the roughness elements $\bar{k}$ [162,167–169]. According to Kuwata et al. [62,167,170,171] the wall-normal distance can be defined as $y_e = \int_0^\delta \varphi dy$, where $\varphi(y) = A_\varphi(y)/A_t$ represents the void fraction, with $A_\varphi(y)$ being the fluid-occupied area at a location y and A_t being the total plane area. In this approach, the effective wall distance y_e is zero at the bottom of the roughness valleys and returns to $y - \bar{k}$ above the maximum roughness peaks, effectively representing the distance from the mean height.

The concept of the wall origin is crucial when calculating the effective boundary layer thickness, δ_e, or effective channel height in channel flows, particularly when roughness occupies a significant portion of the boundary layer or channel. In these situations, δ_e is measured with respect to the wall origin, as shown in Fig. 4, expressed as $\delta_e = \delta - d$ [61,166,172]. When the roughness does not occupy a significant portion of the boundary layer, it can be assumed that $\delta_e \approx \delta$ [46].

Using δ_e instead of δ effectively replaces the roughness with a smooth surface located at $y = \delta_e$, where the no-slip boundary condition is no longer applicable. This concept has been widely employed in the development of roughness extensions for existing Reynolds-Averaged Navier-Stokes (RANS) models [90] to simulate rough-wall turbulent flows without explicitly accounting for the detailed roughness geometry. Examples of such extensions include modifications to the mixing length model [173,174], Spalart-Allmaras model [108,111,175,176,176], k-ε model [177–179], and k-ω model [109,112,114,180–183].

Another method for determining the wall origin for both momentum and heat involves adjusting the velocity and temperature profiles to closely match the slopes of the velocity profile ($1/\kappa$) and the temperature profile ($1/\kappa_\theta$) [4]. Here, $\kappa \approx 0.40$ is the von Kármán constant [184], as discussed in detail by Kadivar et al. [52]. Additionally, $\kappa_\theta \approx 0.459$, which is further explained in section 2.6.

2.4. Averaging methods

In experimental and scale-resolving CFD, such as Direct Numerical Simulations (DNS) and Large Eddy Simulations (LES), flow variables are inherently transient properties. The mean flow parameters are typically evaluated using time-averaging methods to filter out instantaneous fluctuations. However, in rough-wall flows, the surface roughness introduces spatial heterogeneity, causing time-averaged variables to vary in both spanwise and streamwise directions. Therefore, a double-averaging (DA) approach [185] is employed to obtain average flow fields. For any field variable, $\Phi(x, y, z, t)$, DA involves successive applications of the time-averaging and spatial-averaging operators, which are respectively defined as [186]

$$\overline{\Phi}(x,y,z) = \frac{1}{T}\int_{t_0}^{t_0+T} \Phi(x,y,z,t)dt, \tag{3}$$

$$\langle\Phi\rangle(y,t) = \frac{1}{A_\varphi(y)}\iint \Phi(x,y,z,t)dxdz, \tag{4}$$

where $A_\varphi(y)$ represents the fluid-occupied area at y. Below the maximum roughness peak, $A_\varphi = \varphi A_t$, and above this region $A_\varphi = A_t$. DA decomposition leads to

$$\Phi(x,y,z,t), = \underbrace{\overline{\Phi}(y) + \widetilde{\Phi}(x,y,z,t)}_{\overline{\Phi}(x,y,z)} + \Phi'(x,y,z,t). \tag{5}$$

Here $\widetilde{\Phi}(x,y,z,t)$ represents the deviation from the actual or intrinsic component, known as the "form-induced" dispersive component, and $\Phi'(x, y, z, t)$ is known as the turbulent fluctuations component, which is the deviation from the time-average component ($\overline{\Phi}(x,y,z)$).

The DA technique ensures that only the fluid-occupied regions contribute to the mean flow variables. Mean quantities obtained through DA are referred to as "intrinsic averages" [187], in contrast to "superficial averages" where $A_\varphi = A_t$ for the entire flow domain. Both averages are used in the rough flow literature.

2.5. Dimensionless Numbers

The "friction Reynolds number" is defined as $Re_\tau = \delta u_\tau/\nu$, while "bulk flow Reynolds number" is expressed by $Re_b = \delta U_b/\nu$, where U_b is the bulk flow velocity determined by [116]

$$U_b = \frac{1}{\delta}\int_0^\delta U dy \tag{6}$$

The typical spectrum of bulk flow Reynolds number of natural and industrial heat transfer is illustrated in Fig. 5.

Based on the concept of sand-grain roughness (k_s), the "roughness Reynolds number" is defined as [52]

$$k_s^+ = \frac{k_s u_\tau}{\nu}, \tag{7}$$

The typical spectrum roughness Reynolds number is illustrated in Fig. 6.

Nikuradse [151,188] suggested that rough flow can be categorised into three regimes:

(I) The hydraulically smooth regime, where k_s^+ is small and roughness does not affect the momentum transfer. In this regime skin friction and drag coefficient are only a function of the Reynolds number and independent of the roughness.
(II) The transitionally rough regime, where both the Reynolds number and relative roughness influence the skin friction and drag coefficients.
(III) fully rough regime, where the friction drag significantly increases due to the pressure force on the roughness and is independent of the viscous effect (Reynolds number).

The demarcation of these regions and further information on these flow regimes can be found in Kadivar et al. [52].

The problem of convective heat transfer can be reduced to finding the skin friction coefficient and the convective heat transfer coefficient. The skin friction coefficient represents the wall shear stress normalised by the reference velocity U_{ref}, and is [189]

$$C_f \equiv \frac{\tau_w}{\frac{1}{2}\rho {U_{ref}}^2} = \frac{2}{{U_{ref}^+}^2} \tag{8}$$

The heat transfer coefficient is often evaluated using Stanton number (St), defined as [116]

$$St \equiv \frac{q_w}{\rho c_p U_{ref} \Theta_{ref}} = \frac{1}{U_{ref}^+ \Theta_{ref}^+} \tag{9}$$

where $\Theta_{ref} = T_{ref} - T_w$. The reference temperature, T_{ref}, can be chosen as the free stream temperature (T_∞) for external boundary layer flows, and the mixed-mean temperature (T_m) for internal pipe and channel flows [116]. The mixed-mean temperature [190,191] is defined as the temperature when the flow from the heat transfer domain is discharged into a container and thoroughly mixed [66]. Thus, from streamwise convective thermal energy rate (expressed by $(\rho U_b \delta) c_p T_m = \int_0^\delta \rho c_p U T dy$) and Eq. (6), the mixed-mean temperature is obtained as

$$T_m = \frac{\int_0^\delta UTdy}{\int_0^\delta Udy} \tag{10}$$

The mixed-mean temperature is the mass-averaged temperature (or volume-averaged temperature for incompressible flows). It represents the bulk flow temperature, which is different from the arithmetic mean temperature, $T_a = (1/\delta)\int_0^\delta Tdy$ [116].

The frictional drag and heat transfer are functions of the Reynolds number and the roughness length scale, while heat transfer is additionally influenced by the Prandtl number (Pr) defined as [116]

$$Pr = \frac{\nu}{\alpha}, \tag{11}$$

where ν represents the molecular momentum diffusivity and α is the molecular thermal diffusivity. The Prandtl number is a fundamental fluid property that quantifies the relative efficiency of momentum and energy transport through molecular diffusion in the velocity and thermal boundary layers. The Prandtl number spectrum of fluids is illustrated in Fig. 7. When the Prandtl number is close to unity, as seen in gases, energy and momentum transfer by diffusion are equally dominant. However, for $Pr \ll 1$ such as liquid metals, the rate of energy diffusion significantly exceeds the rate of momentum diffusion. Conversely, for $Pr \gg 1$ such as oils, the momentum diffusion rate is dominant.

The Prandtl number significantly influences the growth of velocity and thermal boundary layers. A common relationship for the ratio of momentum to thermal boundary layer thicknesses is $\delta/\delta_t \approx Pr^n$ [192], where $n = 1/2$ for laminar flows [193]. For turbulent flows over smooth walls, $n = 1/3$ is suggested by Kays & Crawford [194] and Bird et al. [195], while some authors have also argued $n = 1/4$ for very large Prandtl numbers [196,197]. According to Fig. 7, $\delta \approx \delta_t$ for gases, $\delta \ll \delta_t$ for liquid metals, and $\delta \gg \delta_t$ for oils.

Various studies [198–202] have explored the analogies between heat (passive scalars) and the velocity field. Additionally, the dissimilarities in scale interactions between the velocity and temperature fields have been discussed in the literature. Kawata and Tsukahara [203] examined some of the differences in small-scale interactions between these fields. The majority of the studies in the field have focused on the case where Pr is between 0.7 (for air) and 1.0. In this particular range of Pr, a close analogy between the temperature and velocity fields exists, while for many engineering and natural fluids, $Pr \ll 1.0$ or $Pr \gg 1.0$. In the context of passive scalar diffusion, the role of the Pr is replaced by the Schmidt number, Sc, which is typically much greater than 1.0 in many practical applications [116]. Under very large or very small Pr (or Sc), the similarity between velocity and heat (or passive scalar) is significantly reduced, making it difficult to predict even the basic properties of the flow.

2.6. Thermal boundary conditions

One of the key differences between momentum and heat transfer is the thermal condition of the surface. In the context of differential equations, there are five types of boundary conditions: Dirichlet, Neumann, Robin, Mixed, and Cauchy, with the first two being the most commonly used. In heat transfer, a uniform wall heat flux (iso-flux) boundary condition represents a Neumann-type boundary, while a uniform wall temperature (iso-temperature) boundary condition represents a Dirichlet-type boundary. Mixed boundary conditions combine elements of both.

Apart from boundary conditions, volumetric heat generation can occur within the fluid through phenomena such as nuclear or chemical reactions. The type of boundary condition and the presence of volumetric heat generation can significantly impact the heat transfer process. The impact of thermal boundary conditions on smooth-wall flow has been studied extensively (see, for example,[204–206]). Avellaneda et al. [204] reported different behaviours in temperature profiles with changes in boundary conditions. Kawamura et al. [206] stated that the turbulent Prandtl number in the near-wall region is independent of the wall boundary condition, whereas, in the central region, it is significantly influenced by the boundary condition. Flageul et al. [205] observed a strong impact of the thermal boundary condition on temperature variance. Some studies [200,207] have shown that, when a constant temperature is imposed at the wall, heat and momentum

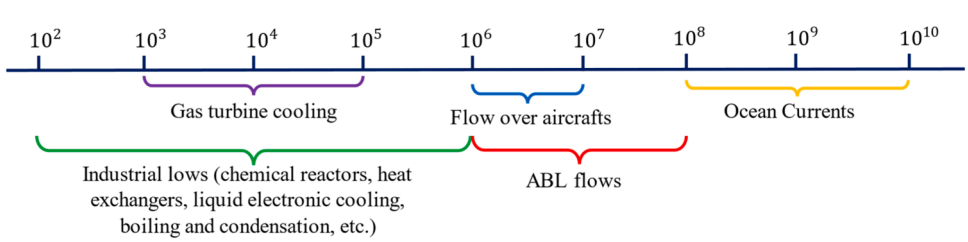

Fig. 5. Typical bulk flow Reynolds number spectrum for natural and industrial heat transfer.

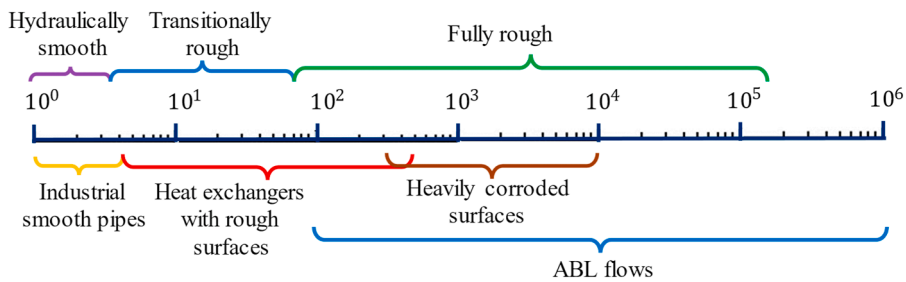

Fig. 6. Typical roughness Reynolds number spectrum for natural and industrial heat transfer.

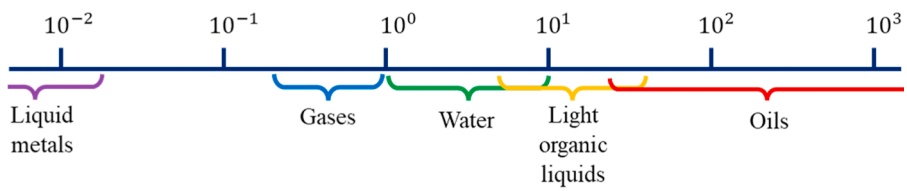

Fig. 7. Typical Prandtl number spectrum fluids [116]

transfer behaviours exhibit close similarity.

Rough-wall flow studies have predominantly used the uniform temperature boundary condition [4,46,55,61,66,73,208,209]. However, the uniform wall heat flux boundary condition is generally regarded as more representative of real-world engineering scenarios, where both the bulk mean and surface temperatures typically increase along the streamwise direction. Due to the use of periodic boundary conditions in CFD studies, implementing a uniform wall heat flux becomes challenging, as it leads to temperature variations in the streamwise direction, making the temperature non-periodic. This complication can be mitigated by using the transformed temperature (Θ) in the energy equation, as Θ remains periodic in the streamwise direction [92,94,210–213]. Some studies [4,66] considered using mixed boundary conditions defined as $T_w = T + \theta_w = T_{ref} + (dT/dx)$. This approach models linear streamwise surface temperature variations similarly to the uniform wall heat flux boundary condition.

Unfortunately, a direct comparison between different thermal boundary conditions in rough-wall flows has not been investigated so far. This comparison would provide insights into the relative advantages and limitations of each approach for simulating heat transfer over rough surfaces in real-world engineering applications.

3. Smooth-wall thermal boundary layer

The behaviour of heat (or more generally passive scalars) in wall-bounded turbulent flows is elaborated in the books by Monin & Yaglom [214], Cebeci & Bradshaw [215], and Kays et al. [116], as well as experimental studies by Gowen & Smith [216], Kader [217], Subramanian & Antonia [218], and Nagano & Tagawa [219]. The turbulent thermal boundary layer borrows various similarities to the corresponding momentum boundary layer. The differences are due to the variety of the different boundary conditions as well as the influence of the molecular and turbulent Prandtl numbers.

In the context of the turbulent boundary layer, apart from the molecular heat flux ($\rho c_p \alpha \partial T/\partial y$) the evaluation of the turbulent heat flux ($\rho c_p \overline{v'T'}$) is of significant importance. Here, v' and T' are the fluctuating components of velocity and temperature, respectively, due to turbulent fluctuations. According to Prandtl's mixing length model, $\overline{v'T'} = \epsilon_H \partial T/\partial y$ [116], where ε_H is the thermal eddy diffusivity. Therefore, the total wall heat flux can be written as [116,220]

$$q_w = -\rho c_p (\alpha + \epsilon_H) \frac{\partial T}{\partial y}; \tag{12}$$

Similarly, the total apparent shear stress can be written as [116]

$$\tau_w = \rho(\nu + \epsilon_M) \frac{\partial U}{\partial y}; \tag{13}$$

where, ϵ_M is the momentum eddy diffusivity. The calculation of thermal and momentum eddy diffusivities is the subject of turbulence modelling. The Prandtl's mixing length theory, suggests $\epsilon_M = \kappa \nu y^+$ [116,220].

As an analogy to the molecular Prandtl number, the turbulent Prandtl number is defined as [116,220]

$$Pr_t = \frac{\epsilon_M}{\epsilon_H} \tag{14}$$

The molecular Prandtl number (Pr) is a constant value determined by the fluid properties; however, Pr_t is a property specific to turbulent conditions which is characterised as the resistance to heat transfer in turbulent flows. A critical examination of the experimental data on Pr_t in engineering flows can be found in Kays [221] and Li [222]. The practicality and limitations of Pr_t were discussed by Kays [221], who explored various approaches—analytical, experimental, and numerical—for calculating Pr_t in different flow types. The Reynolds analogy (see Section 6) provides the most basic model for Pr_t, resulting in a $Pr_t = 1$. However, empirical evidence suggests that Pr_t in the logarithmic region is not equal to 1; instead, it ranges from 0.85 to 0.9, depending on the Reynolds number and the molecular Prandtl number. Some studies [222–224] have shown that, Pr_t decrease to a value close to 0.5 in the wake region of the boundary layer or at the centre of the channel. For a sufficiently high Reynolds number, Pr_t can be assumed to be constant [116,190,207,225–227].

Several studies addressed the impact of the Pr on turbulent flows [213,223,228–234]. Redjem-Saad et al. [223] reported that the turbulent heat flux increases with increasing the Prandtl number leading to enhanced heat transfer. However, the Pr_t remains almost independent of the molecular Prandtl number for $Pr \geq 0.2$.

In the boundary layer, it can be assumed that the streamwise temperature gradient is negligible and temperature is a function of wall-normal coordinate, i.e. $\Theta = \Theta(y)$. Using the concept of the turbulent Prandtl number and mixing length theory, the thermal boundary layer equation can be reduced to [220]

$$\Theta^+ = \int_0^{y^+} \frac{dy^+}{\frac{1}{Pr} + \frac{\kappa y^+}{Pr_t}} \tag{15}$$

The two-layer model can be used to divide the thermal boundary

layer into two layers based on the relative magnitude of $\kappa y^+ / Pr_t$. The first region is located close to the wall, where $\kappa y^+/Pr_t \ll 1/Pr$. This region is known as the "conduction sublayer" (CSL), where molecular conduction dominates heat transfer, and the temperature gradient is primarily influenced by the fluid's thermal diffusivity rather than turbulent mixing. The temperature profile in CSL can be expressed by [116,220]

$$\Theta_\theta^+ = Pry^+,\ y^+ < \delta_\theta^+ \tag{16}$$

where δ_θ^+ is the thickness of the CSL, which depends on Pr. As Pr increases, the thermal diffusivity decreases relative to the viscous effect, resulting in slower conduction compared to convective transport due to momentum, hence reducing δ_θ^+. Various definitions for the thickness of the conduction sublayer have been proposed in the literature (see [213,233,235]). Pirozzoli [229] suggested:

$$\delta_\theta^+ \approx \begin{cases} \dfrac{1}{\kappa_\theta Pr} & Pr \ll 1 \\ \left(\dfrac{{A_\theta}^2}{\kappa_\theta Pr}\right)^{\frac{1}{3}} & Pr \gg 1 \end{cases} \tag{17}$$

where $A_\theta = 19.2$ and $\kappa_\theta = \kappa/Pr_t \approx 0.459$, which is similar to values found in channel flow studies [190,236]. Beyond the CSL, the flow is characterised by a logarithmic profile, where convective transport by turbulent eddies dominates the heat transport [190,206,217,237,238]. For $y^+ > \delta_\theta^+$, the solution yields [194,220]

$$\Theta_S^+ = \frac{1}{\kappa_\theta}\ln y^+ + \beta(Pr),\ y^+ > \delta_t^+ \tag{18}$$

where subscript S represents smooth wall. Here, $\beta(Pr)$ is a Prandtl number-dependent offset function, playing a crucial role in estimating the heat transfer coefficient. The function $\beta(Pr)$ can be determined by [229]

$$\beta(Pr) = \lim_{y^+\to\infty}\left[\Theta^+(y^+) - \frac{1}{\kappa_\theta}\ln y^+\right] \tag{19}$$

In this regard, the most robust framework established to date is the work of Kader and Yaglom [239]. They reviewed some of the proposed correlations for $\beta(Pr)$ derived the empirical correlation

$$\beta(Pr) = 12.5Pr^{2/3} + 2.12\ln Pr - 5.3 \tag{20}$$

Later, Kader [217] revised it to

$$\beta(Pr) = \left(3.85Pr^{1/3} - 1.3\right)^2 + 2.12\ln Pr. \tag{21}$$

Using an integral solution based on the two-layer model, Pirozzoli [229] analytically derived

$$\beta(Pr) = \frac{\sqrt{3}\pi {A_\theta}^{2/3}}{6{\kappa_\theta}^{1/3}}Pr^{2/3} + \frac{1}{\kappa_\theta}\ln Pr - \frac{1}{\kappa_\theta}\ln A_\theta \tag{22}$$

which has an identical structure to Eq. (20). Fig. 8 shows the impact of the increasing Prandtl number on the upward shift of the temperature profile. It also demonstrates the good predictive nature of Eq. (20) and (22), with better results using Pirozzoli's equation.

4. A brief overview of momentum transfer over rough walls

The momentum transfer over roughness has been extensively studied in the literature [51,52,154], thus, this section only provides a short overview this topic. The main impacts of roughness on the momentum transfer can be summarised as (i) increased friction drag [52,151]; (ii) the modification of the near-wall boundary layer and turbulence generation cycle [121,123,154]; (iii) increased anisotropy due to the amplification of wall-normal and spanwise fluctuations [93,240–242].

The impact of surface roughness is reflected in a downward shift of the logarithmic mean velocity profile, known as roughness function (ΔU^+) [52], which is merely a function of k_s^+, supported through various experiments [117,120,155,243–255], DNS [256–261] and LES studies [92–94]. According to Clauser [262], the logarithmic velocity over roughness can be obtained by modifying the smooth-wall log-law.

$$U_R^+ = U_S^+ - \Delta U^+ = \frac{1}{\kappa}\ln(y-d)^+ + C - \Delta U^+, \tag{23}$$

where $C \approx 5.5$ is the smooth-wall intercept [113,184,263]. The subscripts R and S refer to rough and smooth conditions, respectively. Hama [264] proposed the roughness function as

$$\Delta U^+ = \frac{1}{\kappa}\ln k_s^+ + C - B, \tag{24}$$

where B is the roughness constant. In fully rough regime, $B \approx 8.5$ [265], while in the transitionally rough regime, it is a universal function of k_s^+ and is independent of the flow type [266]. The roughness function reflects the momentum deficit due to friction drag by roughness [107,117,246]; thus

$$\Delta U^+ = U_{b,S}^+ - U_{b,R}^+ = \sqrt{\frac{2}{C_{f,S}}} - \sqrt{\frac{2}{C_{f,R}}} \tag{25}$$

This equation is a straightforward method for determining roughness functions in experimental studies. A simpler approach is to define the roughness function as the difference between the outer layer velocity at a matched friction Reynolds number [186]. By definition, the roughness function is the shift of the logarithmic layer due to roughness. Therefore, the most conservative approach is the difference between the mean streamwise velocity between the smooth and rough wall at matched friction Reynolds number [267].

In meteorology and ABL flows, field variables are scaled with a roughness parameter above the canopy, refered to as the "momentum roughness length (y_0)". It can be visualised as the point where the zero velocity axis intercepts the straight line from the logarithmic profile of mean velocity data against elevation within the dynamic sublayer [3]. The value of y_0 is theoretically independent of the flow and depends solely on the characteristics of the surface, including the geometry, size, and arrangement of the roughness elements.

The evaluation of y_0 has received significant attention in the literature, which can be found in reviews by Jiménez [50], Castro [268] and Brutsaert et al. [3]. Raupach et al. [154] provided tabulated data of y_0 for flow over various types of plant canopies, and Raupach [159] developed a simple model for calculating y_0. Some methods for the estimation of y_0 are provided in Brutsaert et al. ([3], chapter 5). Using the momentum roughness length, the velocity profile is described by [3].

$$U_R^+ = \frac{1}{\kappa}\ln\left(\frac{y-d}{y_0}\right) \tag{26}$$

By matching Eq. (23) & (24) with Eq. (26), the relationship between y_0 and k_s can be obtained as

$$k_s/y_0 = \exp(\kappa B) \tag{27}$$

In the case of a fully rough regime where $B \approx 8.5$, Eq. (27) simplifies to $k_s/y_0 \approx 30$ [3].

5. Topological parameters of roughness and their impact on flow and heat transfer

The main challenge in accurately predicting friction drag and heat transfer in practical applications stems from the absence of a reliable method to determine k_s as a function of surface topology alone. As a result, a significant portion of research on rough-wall flows has

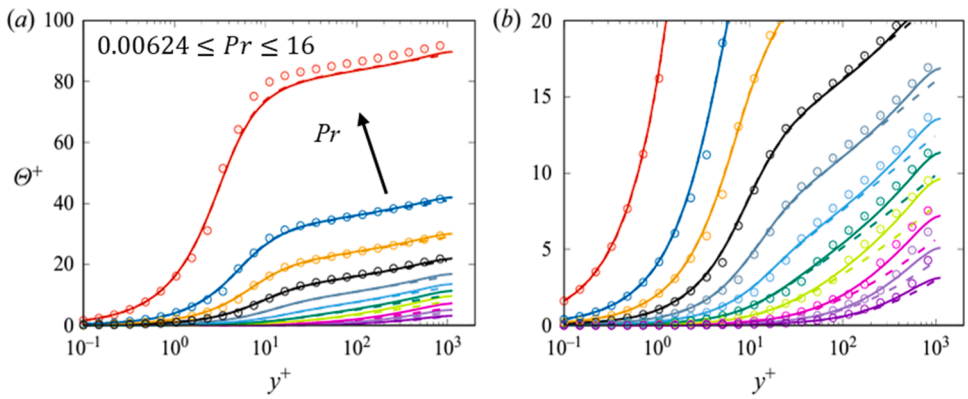

Fig. 8. The impact of the Prandtl number on the temperature profile ($0.00624 \leq Pr \leq 16$) (a) Comparison of mean temperature profiles obtained from DNS of Pirozzoli [229] (solid lines), with the predictions of Eq. (22) (dashed lines) and with Kader's empirical fit Eq. (20) (circles). (b) A magnified view to better observe the behaviour of the low-*Pr* profiles. The figure is adopted from Pirozzoli [229].

concentrated on developing correlations between the topological features of surface roughness. A comprehensive discussion of these parameters and their effects on turbulent flow can be found in Chung et al. [51]. This section provides a brief overview of the key topological parameters of roughness, also referred to as surface properties or roughness statistics.

5.1. Roughness height parameters

The primary characteristic of roughness is its height, which is often considered the most dominant parameter. However, referring to "roughness height" alone can be misleading, as it oversimplifies the complex nature of surface roughness. For regularly shaped roughness geometries, the height of the elements can be directly measured as the height parameter. However, for irregular rough surfaces, defining the roughness height becomes more complex. In these cases, statistical parameters of the roughness height distribution are employed [51,52,269], including the arithmetic average height (S_a), root-mean-square (RMS) height (S_q), mean peak-to-valley height (S_z), and maximum peak-to-valley height (S_t). The roughness parameters can be derived from by a centred height map, $k_c(x,z)$, representing the roughness elevation around the mean value, i.e. $k_c(x,z) = k(x,z) - \overline{k}$. Here, $k(x, z)$ is the height map with respect to $y = 0$ (see Fig. 4), and $\overline{k}$ is the mean height. According to ISO 25178-2 [43],

$$S_a = \frac{1}{l_x l_z} \int_0^{l_x} \int_0^{l_z} |k_c| dxdz \tag{28}$$

$$S_q = \sqrt{\frac{1}{l_x l_z} \int_0^{l_x} \int_0^{l_z} k_c^2 dxdz} \tag{29}$$

where l_x and l_z are the surface dimensions in x and y directions, respectively. Another important height parameter is the maximum peak-to-valley height, expressed as $S_t = S_p + S_v$, where S_p and S_v are the height of the largest roughness peak and the depth of the deepest valley, respectively. Since, S_t can vary for each sampling length, a N-point-average peak-to-valley height (S_z) is instead used, expressed by [52]

$$S_z = \sum_{i=1}^{N} S_{ti} = \sum_{i=1}^{N} (S_{pi} + S_{vi}) \tag{30}$$

$N = 5$ (ISO standard) or $N = 10$ (DIN standard) is typically employed [52]. For regular 2D and 3D roughness $S_z = S_t = k$.

In addition to the primary roughness parameter, other topographical parameters are necessary to capture secondary aspects of roughness that can affect the flow. These include the shape of the height distribution [120,168,257], the spatial arrangement of roughness elements (such as clustering, anisotropy, and regularity/irregularity) [168,259], and the orientation of the rough surface relative to the dominant flow direction [270].

5.2. Roughness spatial (density) parameters

The spatial distribution can be identified by the frontal solidity parameter (λ_f), effective slope (ES), and roughness density (Λ). These parameters represent the available area exposed to form (pressure) drag and can be considered as a measure for roughness density. Several experimental and numerical investigations have indicated that, for a given Reynolds number, the mean velocity is influenced by the roughness density [162,164,271–278]. The frontal solidity is defined as $\lambda_f = A_f/A_t$, where A_f is the projected frontal area of roughness elements and A_t is the total plan area (Fig. 9 (a)). Higher values of λ_f is associated to denser roughness.

The density parameter $\Lambda = \lambda_f^{-1}(A_f/A_w)$, where A_w is the total wetted area, has also been utilised in the literature to characterise roughness density [271,275,279]. The local value of Λ is associated with the local roughness angle α_h [126].

The effective slope is defined as the absolute gradient of roughness elements [162,170,246,280,281], expressed by [51,52]

$$ES = \frac{1}{l_x l_z} \int_0^{l_x} \int_0^{l_z} \left|\frac{\partial k}{\partial x}\right| dxdz \tag{31}$$

and $ES \approx 2\lambda_f$ [257,277,280]. The effective slope can be defined for the streamwise and spanwise direction, although the streamwise direction is particularly significant for rough flow studies. A low ES indicates the roughness elements are either sparsely packed or long-wavelength shallow roughness, while high ES signifies either densely packed or short-wavelength steep roughness. The former is known as k-type roughness, while the latter is referred to as d-type roughness, with further details provided in Kadivar et al. [52].

Among all density parameters (Λ, λ_f, and ES), it appears that only one is sufficient to effectively represent roughness density in correlations [257]. For very sparsely packed or isolated roughness where $\lambda_f \to 0$, the surface drag behaves similarly to a smooth surface, as the exposure of roughness elements to the flow is minimal. For very densely packed roughness where $\lambda_f \to 1$, the roughness elements are tightly spaced and can shelter each other from the flow, as extensively discussed in Macdonald et al. [273], Yang et al. [282], Forooghi et al. [55], and Abu Rowin et al. [68].

The sparse regime is characterised by $\lambda_f \leq 0.15$ ($ES \leq 0.35$), while

flow with $\lambda_f > 0.15$ ($ES > 0.35$) is considered as dense regime [50,281,283]. However, recent studies have reported that this trend varies depending on the skewness of the roughness [64,164]. For constant skewness in the fully rough regime, the transition from sparse to dense regimes occurs at a higher value of λ_f (e.g. greater than 0.21 for block roughness [276]). Within the range of $0.1 < \lambda_f < 0.3$, friction drag is maximised in both fully rough [50,154] and transitionally rough regime [277]. However, for irregular roughness, a monotonic increase in friction drag with increasing roughness density is observed in the fully rough regime [63,170,284].

5.3. Roughness spectral parameters

The main spectral parameters of roughness are plan solidity $\lambda_p = A_p/A_t$, skewness S_{sk}, and kurtosis S_{ku}. A high plan solidity ($\lambda_p \to 1$) indicates a densely packed arrangements of roughness elements, typically associated with negative skewness, while a low plan solidity ($\lambda_p \to 0$) represents a sparsely packed arrangement of roughness elements with positive skewness. The ratio λ_f/λ_p indicates the aspect ratio of roughness elements: cube-like elements have $\lambda_f/\lambda_p = 1$, squat elements have $\lambda_f/\lambda_p < 1$, and tall thin elements have $\lambda_f/\lambda_p > 1$. The spatial parameters in combination with the spectral parameters can enhance the prediction of drag. Studies [273,276,285] have shown that the combined use of plan solidity λ_p and frontal solidity λ_p offers a more comprehensive measure of surface roughness and enhances drag prediction.

Skewness (S_{sk}) is another spectral roughness parameter that measures the asymmetry in surface elevation distribution, defined as [51,52]

$$S_{sk} = \frac{1}{S_q^{\,3}} \int_0^{l_x} \int_0^{l_z} (k_c)^3 dxdz \tag{32}$$

Positively skewed surfaces are typically spikey surfaces with many peaks (Fig. 9 (c)), while negatively skewed surfaces are normally pitted with many valleys (Fig. 9 (c)).

Kurtosis of roughness is a measure of flatness of roughness, defined as [51,52]

$$S_{ku} = \frac{1}{S_q^{\,4}} \int_0^{l_x} \int_0^{l_z} k_c^4 dxdz \tag{33}$$

When $S_{ku} < 3$, the distribution is referred to as platykurtic, indicating the absence of extreme features such as high peaks and low valleys on the surface. Conversely, when $S_{ku} > 3$, the distribution is known as leptokurtic, indicating a prevalence of extreme features. Gaussian roughness is characterised by $S_{sk} = 0$ and $S_{ku} = 3$. Various studies [186,256,258] have shown that, for Gaussian roughness, $k_s \approx S_t$.

According to Chung et al. [51], an accurate evaluation of sand-grain roughness (k_s), and thus drag and heat transfer coefficients, requires including one roughness parameter from each category (height, spatial, and spectral). Skewness, along with effective slope and RMS height (S_q), are previously used in correlating roughness topology to k_s [117,167,286–288]. Compared to Gaussian surfaces, positive skewness increases friction drag, while negative skewness reduces friction drag [248,254,287]. The former is due to increased form drag, while the latter is due to the skimming effect [288], where the flow slides over abundant roughness valleys. Busse & Jelly [267] demonstrated that further increases in skewness beyond certain levels do not significantly affect the drag produced by the surface. Additionally, while surfaces with extreme features (such as high kurtosis) can locally impact flow fields, these deviations affect only a small portion of the flow due to the relative sparseness of such extreme features.

5.4. Impact of roughness topology on heat transfer

Similar to friction drag, heat transfer is significantly influenced by roughness topology; however, our understanding of this effect remains less developed, primarily due to a lack of systematic studies. Heat transfer is typically greater on the windward surface of roughness elements [61,92,94], where unmixed fluid with high-temperature gradients interacts. In contrast, the leeward surface experiences well-mixed, low-gradient fluid. In ABL flows, a highly turbulent region formed behind leeward face of mountains leads to the formation of a natural "banner clouds" around isolated mountain peaks [289,290].

Fig. 10 illustrates the instantaneous and average flow from the DNS data of Peeters & Sandham [61]. Flow impinges on the windward slope, and recirculation forms in the wake region behind the roughness elements. The strength of the recirculation can increase with increasing k_s, resulting in a momentum deficit. Increasing k_s also bring more fresh fluid from the bulk flow region into the roughness layer. The fresh fluid typically impinges on a windward slope (Fig. 10, region B), forming a thin thermal boundary layer, extending both in the streamwise direction, influenced by the mean streamwise momentum, and in the spanwise direction, influenced by the recirculation zone and the adverse pressure gradient. It terminates once it comes under the influence of the mean streamwise momentum again (Fig. 10, region C).

The roughness density, specifically the frontal solidity λ_f, controls the flow impingement mechanism on the windward surface [285]. Similar to friction drag, heat transfer indicates a significant increase within the range of $0.1 < \lambda_f < 0.3$ [55,64]. For $\lambda_f \geq 0.3$ sheltering effect occurs, which has been extensively studied in the context of friction drag [51,265,273,278,282,291,292]. The impact of λ_f and sheltering effect on heat transfer has recently attracted research attention [55,63,64,68,293,294]. The flow sheltering effect generally decreases drag in dense regimes, as shown by multiple studies reviewed by Chung et al. [51]. Forooghi et al. [55,164] suggested that the sheltering effect similarly impacts the heat transfer coefficient due to the weaker temperature gradients that forms in the low-speed flow within the sheltered wake regions. Kuwata [63] observed an increase in heat transfer with ES, regardless of S_{sk}, consistent with the trend of friction drag.

6. Heat and momentum transfer analogy

DNS studies on smooth-wall flows [203,210,225,237,295] have shown that scalar (temperature) fluctuations and wall-normal fluxes are

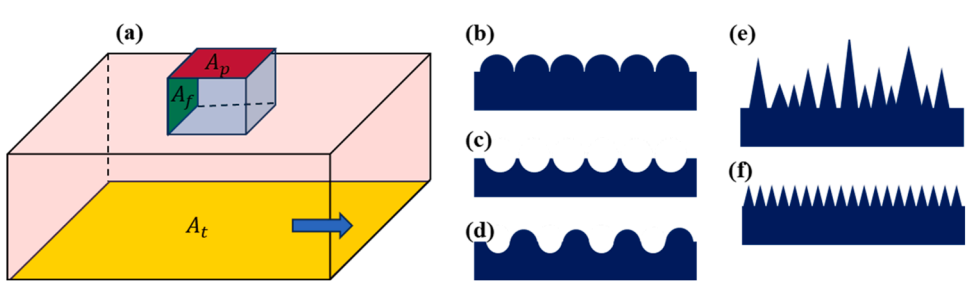

Fig. 9. (a) Schematic representation of projected surface area (A_f), plane surface area (A_p), and total plane area (A_t), used to define frontal and plane solidity. (b)-(f) Schematic of positive skewness (b), negative skewness (c), Gaussian roughness (d), leptokurtic (e), and Platykurtic (f).

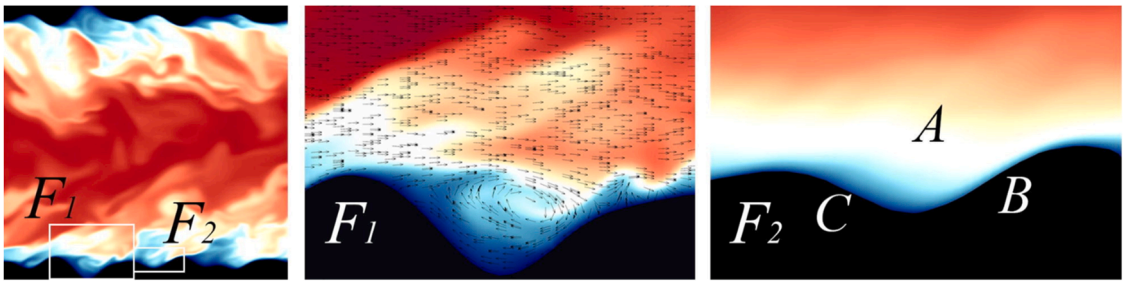

Fig. 10. Left: a segment of the simulated channel displaying instantaneous temperature contours. Middle: instantaneous velocity vectors near a recirculation zone from the frame F_1, with velocity vectors not to scale. Right: time-averaged temperature contours near a recirculation zone from the frame F_2. The figure is adopted from Peeters & Sandham [61].

closely related to streamwise velocity fluctuations and wall-normal stress, especially near the wall. Both kinetic energy and scalar variance spectra exhibit reasonable similarity in the inner and outer regions, as well as in streamwise and spanwise directions. These studies have also analysed the budgets of scalar fluctuations, noting their similarity to those of streamwise velocity fluctuations. For flows with $Pr \approx 1$ at sufficiently high Reynolds number, the ratio between the characteristic time scales of velocity and scalar fluctuations remains relatively constant throughout most of the boundary layer. However, differences can arise between large-scale scalar structures and their streamwise velocity counterparts, with scalar structures often exhibiting a larger spanwise extent and a steeper inclination angle.

The Reynolds analogy explains that the heat and momentum transfer largely depend on the same turbulent scales (eddies), which inspired several researchers to find relations between St and C_f. Experimental heat transfer over $0.6 < Pr < 60$ supports the empirical formula suggested by Colburn [296] for smooth-wall flows

$$\frac{2St}{C_f} = Pr^{-2/3} \tag{34}$$

This relation is referred to as the Colburn analogy, also known as Chilton–Colburn analogy. In the special case where $Pr = 1$, Colburn's analogy simplifies to $2St/C_f = 1$, which is known as the Reynolds analogy, where $RA = 2St/C_f$ is referred to as the Reynolds analogy factor. A direct result of Reynolds analogy is $Pr_t = 1$ [220], which suggests that each eddy has an equal propensity to transfer heat as it does to transfer momentum in the direction normal to the wall. This equivalency means that convective heat transfer corresponds to the same amount of heat transfer (gain) as the momentum deficit (loss). However, even at $Pr_t = 1$, the Reynolds analogy does not universally apply to all physical mechanisms.

At the smallest turbulence scales, heat diffusion is always dominated by molecular conduction, whereas momentum diffusion results from chaotic motions and interactions among turbulent eddies. Kays et al. [116] suggested that these differences stem from the lack of analogy between heat transfer mechanisms and pressure mechanisms for momentum transfer. Dipprey & Sabersky [297] also demonstrated that the Colburn analogy, which incorporates the impact of the Prandtl number, can underpredict the Stanton number (e.g. 2.5 % below the measured value at $Pr = 6$).

Employing the "two-layer model" of Prandtl [298], for the smooth-wall boundary layer, it can be shown that [299]

$$RA_S = \frac{1}{1 + 5\sqrt{\frac{C_{f,S}}{2}}\,(Pr - 1)} \tag{35}$$

where subscript S signifies the smooth wall. This relation is known as the Prandtl analogy, and for $Pr = 1$, it is identical to the Reynolds analogy. Von Karman [184] further improved the Prandtl anology by including the buffer layer, between the viscous sublayer and the fully turbulent core, where molecular and turbulent diffusivities are of similar magnitude. The von Karman analogy yields

$$RA_S = \frac{1}{1 + 5\sqrt{\frac{C_{f,S}}{2}}\left[Pr - 1 + \ln\left(\frac{5Pr+1}{6}\right)\right]} \tag{36}$$

For $Pr = 1.0$, the von Karman analogy is also identical to the Reynolds analogy. A significant limitation of these analogies is their reduced effectiveness for $Pr \ll 1$, such as molten metals. For these low-Prandtl-number flows, Martinelli [300] proposed

$$RA_S = \frac{1}{5\sqrt{\frac{C_{f,S}}{2}}\left[Pr + \ln(5Pr + 1) + 0.5\ln\left(\frac{Re}{60}\sqrt{\frac{C_{f,S}}{2}}\right)\right]} \tag{37}$$

The investigation and verification of heat and momentum transfer analogy, under various flow conditions, have been the subject of various experimental and analytical studies [301–316].

In cases where there is a variation in the pressure gradient, the Reynolds analogy is affected because heat transfer is generally less sensitive to pressure gradient changes compared to momentum transfer [317]. Roughness induces substantial pressure gradient variations, causing significant alterations to the Reynolds analogy. At $Pr = 1$, the only difference between the governing equations of heat and momentum is the pressure gradient. Therefore, it is expected that, for rough-wall flows, the RA factor will be a function of Pr and k_s^+. It can be analytically demonstrated that [63,68,297,318]

$$RA_R = \frac{1}{Pr_t + \sqrt{\frac{C_{f,R}}{2}}\,St_k^{-1}} \tag{38}$$

where St_k^{-1} is the inverse of the roughness Stanton number, detaily discussed in Section 7.1. Table 1 presents various empirical RA formulas for rough-wall flows.

7. Rough-wall heat transfer models

Due to the chaotic nature of turbulence, no unified theory exists to fully describe turbulent flow. As a result, rough-wall heat transfer is typically characterised using empirical and phenomenological models. Empirical models are formulated based on experimental data, which limits their applicability to specific flows, roughness types, and the ranges of Reynolds and Prandtl numbers. In contrast, phenomenological models are grounded in physical observations, scaling laws, and established hypotheses, providing a broader and more conceptual approach.

The logarithmic behaviour of a rough-wall thermal boundary layer is governed by dimensional parameters, including y^+, k_s^+, and Pr, i.e. $\Theta^+ = f\left(y^+, k_s^+, Pr\right)$. Here, k_s^+ is a function of Re_τ, roughness height, as well as the shape of roughness elements and their distribution [51]. The impact of rough walls on heat transfer is captured in the logarithmic intercepts

Table 1
Correlations proposed for the Reynolds analogy factor (RA) in the fully rough regime.

Author	correlation	Flow and roughness
Dipprey & Sabersky [297]	$RA_R = \dfrac{1}{1+\sqrt{\frac{C_{f,R}}{2}}\left[5.19\left(k_s^+\right)^{0.20}Pr^{0.44}-8.5\right]}$	Fully rough, Sand-grain, $1.20 \le Pr \le 5.94$
Kays & Crawford [194]	$RA_R = \dfrac{1}{Pr_t+\sqrt{\frac{C_{f,R}}{2}}\left[1.25\left(k_s^+\right)^{0.20}Pr^{0.44}\right]}$	Fully rough, Sand-grain, $Pr = 0.71$
Wassel & Mills [318]	$RA_R = \dfrac{1}{Pr_t+\sqrt{\frac{C_{f,R}}{2}}\left[4.8\left(k_s^+\right)^{0.20}Pr^{0.44}-7.65\right]}$	Fully rough, Sand-grain, $Pr = 0.71$
Bons [125]	$RA_R = \dfrac{1}{1+\sqrt{\frac{C_{f,R}}{2}}\left[5.96\left(k_s^+\right)^{0.20}Pr^{0.44}-8.5\right]}$	Fully rough, Real gas turbine degraded surface, $Pr = 0.71$
Webb et al. [319]	$RA_R = \dfrac{1}{1+\sqrt{\frac{C_{f,R}}{2}}\left[4.5\left(h^+\right)^{0.28}Pr^{0.57}-0.95\left(\frac{p}{h}\right)^{0.53}\right]}$	Fully rough, Repeated spanwise square rib with height h and pitch p $Pr = 0.71, 5.10, 21.7$
Belnap et al. [320]	$RA_R = \dfrac{1}{Pr_t+\left(\frac{C_{f,R}}{2}\right)^{0.1}\left(St_k^{-1}-1\right)}$	Transitionally rough, Real gas turbine blades, $Pr = 0.71$
Nunner [321]	$RA_R = \dfrac{1}{1+1.15Re^{-\frac{1}{8}}Pr^{-\frac{1}{6}}\left(Pr\frac{C_{f,R}}{C_{f,S}}-1\right)}$	-
Park & Chung [322]	$RA_R = \dfrac{1}{1+\sqrt{\frac{C_{f,R}}{2}}\left[5.38\left(k_s^+\right)^{0.26}Pr^{0.57}-7.7\left(k_s^+\right)^{-0.20}\right]}$	Transitionally rough, Repeated spanwise riblets, $Pr = 2014$
Park & Chung [322]	$RA_R = \dfrac{1}{1+\sqrt{\frac{C_{f,R}}{2}}\left[5.38\left(k_s^+\right)^{0.26}Pr^{0.57}-3.19\right]}$	Fully rough, Repeated spanwise riblets, $Pr = 2014$

that are added to the smooth wall log-law, as an analogy to ΔU^+. Various researchers [191,297,323,324] have attempted to obtain a similar term for heat transfer and scale it with dimensionless flow parameter in the form of $\left(k_s^+\right)^{-p}Pr^{-m}$. In smooth-wall flows, the exponents $m = 1/2$ for $Pr \le 1$ and $m = 2/3$ for $Pr \gg 1$ have been widely supported by Prandtl–-Blasius-type arguments for laminar boundary layers [113,116] and classical natural convection [325–329]. This is based on the assumption of cubically increasing eddy diffusivities in the viscous-conductive sublayer region of turbulent flow [2,217]. Colburn analogy suggests $m = 2/3$, while other authors [115,196,197] have proposed $m = 3/4$ for $Pr \gg 1$ in smooth-wall flows. MacDonald et al. [66] proposed that local flow over rough walls might be explained using a similar relationship found in smooth walls.

7.1. Cavity vortex hypothesis

This was proposed by Owen & Thomson [191] in 1962, and it is probably the oldest rough-wall heat transfer model. A similar approach was published by Dipprey & Sabersky [297] in 1963 taken from Dipprey's PhD thesis [330] in 1961 at the California Institute of Technology. The term "cavity vortex hypothesis" was suggested by Dipprey & Sabersky [297]. This hypothesis employs the observations of turbulent flow structure close to the wall and leverages the analogy between heat and momentum.

The cavity vortex (CV) hypothesis assumes a fully rough flow regime, where drag is independent of viscous effects, and the mean flow pattern remains unaffected by viscosity. Another key assumption is that the roughness elements are closely spaced, causing a sharp decrease in shear stress and flow velocity as the flow approaches the roughness peaks. As a result, the mean flow does not penetrate between the roughness elements, treating the roughness as a sublayer of highly sheared flow. Within this sublayer, the streamwise velocity changes by an amount on the order of u_τ over a distance of k_s.

The CV hypothesis relies on the concept of Coherent Flow Structures (CFS), schematically illustrated in Fig. 11. Modern simulations and experiments have demonstrated that the turbulent structure in shear flow is governed by CFSs. Two basic forms of CFSs include horseshoe and hairpin vortices [331,332]. Horseshoe vortices are counter-rotating vortices around the roughness elements, forming a shape resembling a horseshoe and creating a complex flow structure around the element. Hairpin vortices are elongated, U-shaped vortices that resemble a hairpin in shape and consist of a pair of counter-rotating vortices connected by a curved flow structure. CFSs contain most of the turbulent energy and are typically found in the productive subrange of the flow. These coherent structures are also responsible for the generation of velocity streaks, which manifest as alternating regions of high and low-speed fluid. Streaks play a crucial role in modulating the transport of heat and momentum in the boundary layer.

CFSs draw fluid into the valleys and spaces between roughness elements, scour these regions, and then mix the fluid back into the main flow at the level of the roughness peaks. According to the CV hypothesis, this scouring action is considered the primary mechanism for heat transfer at the wall. The horseshoe eddies extend parallel to the wall and move between the roughness elements, facilitating heat transfer between the surface and the mainstream fluid. They acquire vorticity from the streamwise flow as it moves away from the roughness valleys and experience frictional resistance at the solid surface, where heat is transferred via conduction. The heat is then transferred to the fluid near and above the roughness peaks through the convective motion of the horseshoe eddies.

The horseshoe eddies influence the fluid flow, with a thickness approximately equal to height of roughness layer. Since the characteristic thickness of the roughness sublayer is k_s and the characteristic velocity is u_τ, the heat transfer rate across the unit area of the sublayer is described by [297]

$$q_w = \rho c_p u_\tau (T_w - T_k) St_k \tag{39}$$

where T_w is the wall temperature, T_k is the temperature at the outer edge

of the roughness sublayer. Here, St_k is referred to as the "roughness Stanton number" [116,333–335] or the "sublayer Stanton number" [191] (also represented by B in meteorology and ABL flows). St_k represents the thermal resistance to the heat flow due to finite temperature drop in the roughness cavities as well as between and around roughness elements [297]. Based on the scaling analysis, Owen & Thomson [191] proposed that a possible form of St_k can be

$$St_k = \frac{1}{\alpha_k}\left(k_s^+\right)^{-p} Pr^{-m} \tag{40}$$

where α_k is a constant value dependent merely on the roughness type. The exponents p and m depend on both molecular and turbulent diffusivities in the sublayer. Although Eq. (40) was developed using experimental data from internal flows, it has also been tested for airflow external boundary layer [336]. The values for p and m remain a subject of debate in the context of convective heat transfer over rough surfaces [4,337,338]. Various values, ranging $p = 0.20 - 0.5$ and $m = 0.44 - 0.8$, have been proposed for $Pr \geq 1$, based on data from laboratory experiments and meteorological observations on different roughness topologies [3,335]. Owen & Thomson [191] employed a scaling analysis method similar to the Prandtl–Blasius laminar boundary layer theory [113] and determined $p = 1/2$ and $m = 3/4$. This scaling approach assumes that the local viscous sublayer covers the roughness layer. The schematic of the mechanisms in CV hypothesis is illustrated in Fig. 12, which also summarises the scaling analysis.

Owen & Thomson [191] fitted Eq. (40) to experimental data for various roughness types, including Dipprey & Sabersky data [297,330], covering a Prandtl number range of 0.7 to 5.9. They determined values of $p = 0.45$ and $m = 0.80$, which approximately matched their scaling data. They also observed that α_k is ranging from 0.45 to 0.70, depending on the roughness topology, and suggested the weighted average of the range ($\alpha_k = 0.52$) provides the best approximate estimation in the absence of measurement data.

Dipprey & Sabersky [297,330] proposed an alternative set of exponents and coefficients based on experimental data for heat transfer in pipes roughened by sand indentation: $\alpha_k = 5.19$, $p = 0.20$ and $m = 0.44$. These values are notably different from the values proposed by Owen & Thomson [191], with their exponents being approximately half and their coefficient ten times larger. This difference may be attributed to the different velocity scales used in these studies. Owen & Thomson [191] employed u_τ as the velocity scale in the sublayer; however, Dipprey & Sabersky [297,330] utilised velocity at the level of the roughness peaks (tips), expressed as $u_g = Au_\tau$. They also assumed $A = 8.49$ for the fully rough regime to form a certain limiting case. This can explain the higher value of α_k obtained in their fitting process. It is worth mentioning that the value of A is nearly identical to the roughness constant (B in Eq. (24)) in the fully rough regime. As shown later (in section 8), $A = Pr_t B$, and when $Pr_t = 1$, $A = B$.

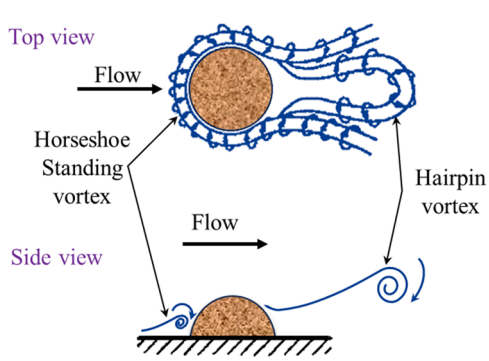

Fig. 11. Schematic illustration of the horseshoe and hairpin vortices over a hemispherical roughness, re-generated from Acarlar & Smith [331] by permission of Cambridge University Press.

Using u_g as the velocity scale, Dipprey & Sabersky defined a different Stanton number near roughness, referred to as the "cavity Stanton number", expressed by $St_c = q_w/\left[\rho c_p u_g (T_w - T_c)\right]$. A closer comparison reveals that $St_c \propto f\,(Ak_s^+,\ Pr)$ and $St_k \propto f\,(k_s^+,\ Pr)$; however, these quantities should be equal over the rough surface. To address this, Dipprey and Sabersky [297] defined a g_k function, known as "roughness similarity function", which resembles the inverse of the St_k, expressed as

$$g_k = \alpha_c\left(k_s^+\right)^p Pr^m \tag{41}$$

Later in section 8, we demonstrated that $St_k^{-1} = g_k - Pr_t B$. Table 2 shows a list of correlations proposed for St_k^{-1} and g_k found in the literature.

Fig. 13 compares some of St_k^{-1} correlations with the DNS data of Kuwata [63] for $Pr = 1.0$, LES data of Garg et al. [92,94] for $Pr = 0.7$, and experimental data of Dipprey & Sabersky [297] for $Pr = 1.2$. There is a significant discrepancy among the predictions of the correlations, which may be due to the utilisation of different roughness topologies in each study. This suggests a need for a unified theory that employs universal flow scales (velocity and length scale) that considers a broader range of roughness topological parameters, including spatial (density) and spectral parameters. This topic is further discussed in Section 7.3.

Wang et al. [353] reported that the correlation proposed by Dipprey & Sabersky [297] aligns well with their experimental measurements. However, Barlow et al. [354] noted that the correlation agrees with the experimental data only when the coefficient is set to 1.0. The correlation by Owen & Thomson [191] has been acknowledged for providing an acceptable approximation for irregular roughness and has been therefore used in ice accretion studies [110,355–358].

In the CV hypothesis, calculating the thermal resistance in the sublayer, represented by St_k^{-1} (equivalent to $T_w - T_k$), necessitates a detailed model of fluid motion within the gaps between roughness elements that is both physically accurate and mathematically feasible. However, developing such a model has proven to be an exceptionally challenging, if not impossible. This difficulty likely explains why Owen & Thomson [191] and Dipprey & Sabersky [297] resorted to scaling analysis and empirical fitting to derive St_k^{-1}.

Jayatilleke [333] and Millionshchikov [359] refined the CV model by incorporating specific momentum and heat eddy viscosities. Despite these efforts, the model has not garnered significant attention in the literature. Yaglom & Kader [346] critiqued the appropriateness of using k_s and u_τ as the characteristic length and velocity scales, respectively, and proposed an alternative method for estimating St_k^{-1}, without necessitating detailed knowledge of the flow motion in the sublayer. Drawing from smooth-wall considerations [239,360], they posited that the eddy diffusivities should decrease by y^3. Through scaling arguments, they demonstrated that when $Pr \approx 1.0$ their model aligns with the results of Owen & Thomson [191]. However, for $Pr \gg 1.0$, they derived $St_k^{-1} = \left(k_s^+\right)^{\frac{1}{2}}\left[b'_1 Pr^{\frac{2}{3}} + b'_2\right]$. In a subsequently study [361], they adjusted the model by introducing an additional fitting parameter, γ, to replace the 1/2 exponent, refining the expression as follows

$$St_k^{-1} = \left(k_s^+\right)^{\gamma}\left[b'_1 Pr^{\frac{2}{3}} + b'_2\right] \tag{42}$$

where $\gamma = 1/2$ for 3D and $\gamma = 1/4$ for 2D roughness. Yaglom & Kader [346] validated their expression through comparisons with roughness configurations involving closely packed spheres, cylinders, and the wave-form roughness elements of Chamberlain [362].

7.2. Surface Renewal (Kolmogorov–Brutsaert) theory

Surface renewal theory [3,337,338,363,364] was initially proposed by Brutsaert [323,324] by utilising the Kolmogorov's energy-cascading phenomenology of turbulent eddies to describe local heat transfer and

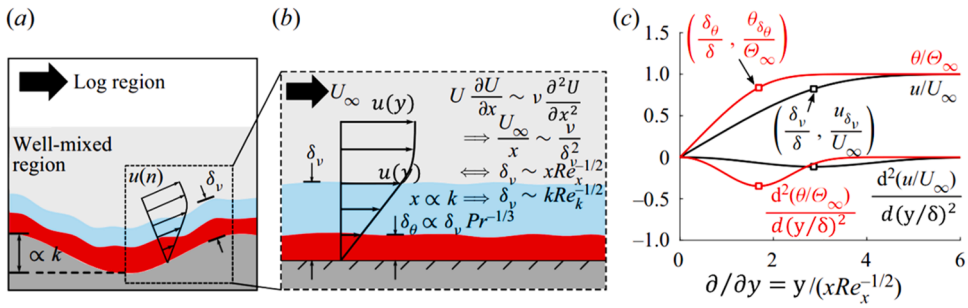

Fig. 12. (a) Schematic of the fully rough model representing the Prandtl–Blasius-type laminar boundary layers cover the entire rough surface. (b) Schematic illustration of the case where the $Pr \approx 1$, resulting in the thermal boundary layer thickness $\delta_\theta \propto \delta_v Pr^{m-1} = \delta_v Pr^{1/3}$, where δ_θ and δ_v are the thickness of the CSL and VSL, respectively. (c) Example solutions from Prandtl–Blasius theory depict velocity (black) and temperature (red) profiles for $Pr = 5.0$. The linear VSL and CSL regions, δ_θ and δ_v, respectively, are indicated by the minima of their second derivatives (marked by square markers). The original figure is adopted from Zhong et al. [4] and modified by the authors.

Table 2
Alternative correlations proposed for St_k^{-1} and g_k for fully rough regime

Author	Roughness type	correlation	k_s^+ range	Pr range
Primenta et al. [339]	Closely packed spheres	$St_k^{-1} = 1.25(k_s^+)^{0.20}Pr^{0.44}$	$k_s^+ \geq 70$	$Pr = 0.7$
Owen & Thomson [191]	Various 3D irregular and regular roughness	$St_k^{-1} = 0.52(k_s^+)^{0.45}Pr^{0.8}$	$k_s^+ \geq 70$	$0.72 \leq Pr \leq 5.9$
Jayatilleke [333]	Irregular roughness	$St_k^{-1} = 0.63(k_s^+)^{0.359}Pr^{0.695}$	-	$0.7 \leq Pr \leq 100$
Dipprey & Sabersky [297]	Sand-grain	$g_k = 5.19(k_s^+)^{0.20}Pr^{0.44}$ $St_k^{-1} = 5.19(k_s^+)^{0.20}Pr^{0.44} - 8.48$	$k_s^+ \geq 65$	$1.20 \leq Pr \leq 5.94$
Wassel & Mills [318]	Sand-grain	$g_k = 4.8(k_s^+)^{0.20}Pr^{0.44}$	-	$1.20 \leq Pr \leq 5.94$
	Repeated spanwise riblets with height k	$g_k = 4.3(k^+)^{0.22}Pr^{0.57}$	-	$Pr = 0.71$
Bons [125]	Real gas turbine degraded surface	$g_k = 5.96(k_s^+)^{0.20}Pr^{0.44}$ $St_k^{-1} = 5.96(k_s^+)^{0.20}Pr^{0.44} - 8.5$	-	$Pr = 0.71$
Ligrani et al. [340–343]	Developing flow over spherical/hemispherical roughness	$g_k = C(k_s^+)^{0.20}Pr^{0.44}$	-	$Pr = 0.71$
		$C = 1.0$ for the transitionally rough regime, and $C = 2.86$ for uniformly distributed spheres for closely packed spheres. Léon et al. [344] proposed $C = 1.78$ for uniformly distributed hemispherical.		
Sheriff & Gumley [345]	Repeated spanwise rods	$g_k = 5.37(k_s^+)^{0.199} + 2.03$	$k_s^+ \geq 50$	-
Webb et al. [319]	Repeated spanwise square rib with height k and pitch p	$g_k = 4.50(k^+)^{0.28}Pr^{0.57}$ $St_k^{-1} = 4.5(k^+)^{0.28}Pr^{0.57} - 0.95\left(\frac{p}{k}\right)^{0.53}$	$k_s^+ \geq 25$	$Pr = 0.71, 5.10, 21.76$
Yaglom & Kader [346]	Closely spanwise pack sphere, cylinders, and wave-from repeated ribs	$g_k = 0.55(k_s^+)^{\gamma}\left[Pr^{\frac{2}{3}} - 0.2\right] + 9.5$ $\gamma = \frac{1}{2}$ for 3D and $\gamma = \frac{1}{4}$ for 2D roughness.	$k_s^+ \geq 70$	$1.20 \leq Pr \leq 5.94$
Dalle Donne & Meyer [347]	Repeated trapezoidal ribs	$g_k = 5.8(k_s^+)^{0.20}Pr^{0.44}\left(\frac{T_w}{T_E}\right)^{0.2}$ $(T_w/T_E)^{0.2}$ was introduced based on the recommendation of Dalle Donne & Meerwald [348], for large variations of the gas physical properties due to the large temperature differences.	$k_s^+ \geq 70$	$Pr \approx 0.7$
Hundia [349]	Annular channel, repeated ribs with height k	$g_k = 4.5(k^+)^{0.24}Pr^{0.44}$	$k^+ \geq 2$	-
Sethumadhavan & Rao [350]	Tube with helical-wire-coil insert with a helix angle (α) and height k	$g_k = 8.6(\tan\alpha)^{-0.18}(k^+)^{0.13}Pr^{0.55}$	$k^+ > 30$	Pure water and 50% water + 50% glycerol
Park & Chung [322]	Repeated spanwise square ribs	$g_k = 5.38(k_s^+)^{0.26}Pr^{0.57}$ $St_k^{-1} = 5.38(k_s^+)^{0.26}Pr^{0.57} - 7.7(k_s^+)^{-0.20}$ $St_k^{-1} = 5.38(k_s^+)^{0.26}Pr^{0.57} - 3.19$	$82.64Pr^{-0.335} > k_s^+$ $82.64Pr^{-0.335} < k_s^+ < 70$ $70 < k_s^+$	$Pr = 2014$
Dawson & Trass [351]	Mass transfer over V-shaped grooves	$g_k = 5.5(k_s^+)^{0.25}Sc^{0.58}$	-	$390 < Sc < 4600$

Table 3
Summary of the recent DNS and LES studies on $\Delta\Theta^+$ [4,58,61–63,66,68,92,94,352].

Author	Method	Roughness	k_s^+	Pr
MacDonald et al. [66]	DNS	Sinusoidal	44 - 379	0.7
Peeters & Sandham [61]	DNS	Irregular Gaussian	13 - 101	1.0
Zhong et al. [4]	DNS Minimal channel	Sinusoidal	18 - 300	0.5
				0.7
				1.0
				2.0
Kuwata [63]	DNS	Irregular $ES = 0.32$ ($\lambda_f = 0.16$)	71 - 177	1.0
		Irregular $ES = 0.64$ ($\lambda_f = 0.32$)		
Kuwata [62]	DNS	Irregular Gaussian	32 - 690	1.0
Garg et al. [92]	LES	Irregular	17 - 406	0.7
Garg et al. [94]	LES	Irregular ($S_{sk} = +0.424$)	1 - 536	0.7
		Irregular ($S_{sk} = -0.424$)		0.7
Hantsis & Piomelli [58]	DNS	Irregular	70 - 485	2.0
De Maio [352]	DNS	Irregular	23 - 300	0.7
Rowin et al. [68]	DNS minimal channels	Sinusoidal $\lambda_f = 0.09$	17 - 327	0.7
		Sinusoidal $\lambda_f = 0.15$		0.7
		Sinusoidal $\lambda_f = 0.36$		0.7

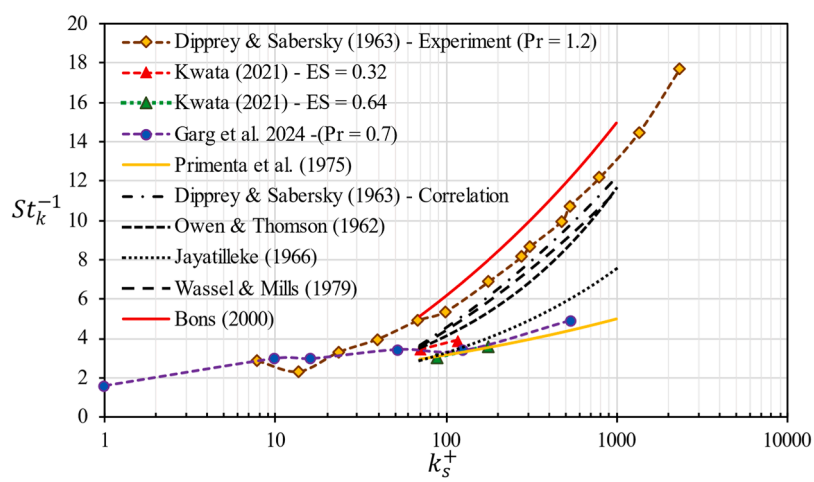

Fig. 13. Comparison of various correlations in calculating the inverse roughness Stanton number. The results of correlations are compared with the DNS data of Kuwata [63] for (Pr =1.0), LES data of Garg et al. [92,94] for (Pr = 0.7), and experimental data of Dipprey & Sabersky [297] (Pr = 1.2). The correlations are evaluated at Pr = 1.0

evaporation (mass transfer) in boundary layer flows. Thus, this theory is also known as the Kolmogorov–Brutsaert (KB) theory. A key distinction between the KB theory and the CV hypothesis lies in their emphasis on surface conditions. The KB theory was initially developed for meteorological and ABL flows and has been extensively studied and applied in the field to calculate surface fluxes of momentum, heat, and mass over both smooth and rough surfaces [337,365–368].

As shown in Fig. 14 (a), Brutsaert's model posits that, in the fully rough regime, the roughness induces eddies with a size proportional to the roughness height k, thereby creating a new scale separation between the roughness-scale eddies and the Kolmogorov eddies (with size $\eta = (\nu^3/\varepsilon)^{1/4}$), scouring the surface area. A uniform, well-mixed temperature, termed the interfacial temperature, Θ_i, dominates in the region from the CSL to an interfacial height, y_i. Above this height, the temperature profile transitions to a logarithmic form [191,297,346]. This roughness energy cascading is believed to commence at the beginning of the logarithmic layer ($y_i \propto k$). Roughness provides a calm (sheltered and windless) environment for Kolmogorov eddies, which initially carry the interfacial temperature Θ_i from the well-mixed region, to span the CSL and facilitate a surface renewal process. During their contact time t, these stagnant sheltered Kolmogorov eddies transfer their temperature onto the surface in this calm environment, which lacks VSL. This renewal process is analogous to the repeated random cycles of "bursting" and "ejection" of fluid, followed by an "inrush" of fresh fluid, a phenomenon observed in various flow visualisation studies [369–371]

The energy cascading can be characterised by the turbulent energy dissipation rate $\varepsilon \approx {u_\tau}^3/\kappa y_i \propto {u_\tau}^3/\kappa k$, which continues down to the Kolmogorov eddies. These eddies have a size $\eta = (\nu^3/\varepsilon)^{1/4} \propto \left[\kappa y_i (\nu/u_\tau)^3\right]^{1/4}$ and a lifetime of $t_\eta = (\nu/\varepsilon)^{1/2} \propto (\nu\kappa y_i/{u_\tau}^3)^{1/2}$ [323,324].

In the fully rough regimes, the disrupted VSL eliminates the existence of thermal advection. As a result, the temperature distribution is governed by unsteady diffusion in the local wall-normal direction n, expressed by [323,324]

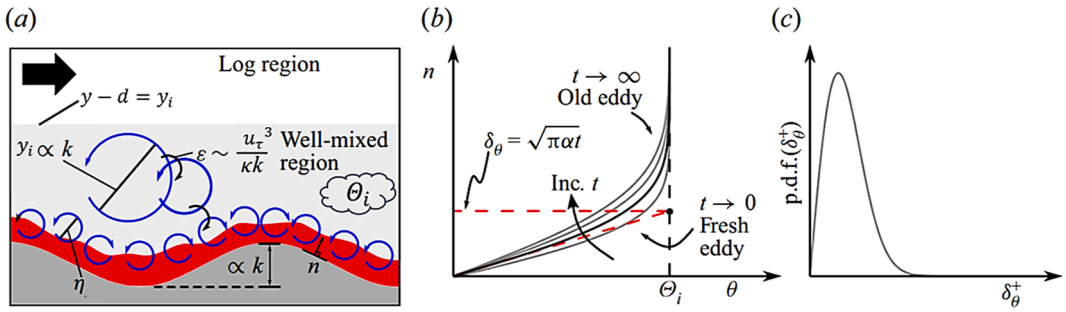

Fig. 14. (a) Schematic of the surface renewal (Kolmogorov–Brutsaert) model based on Kolmogorov energy cascading phenomenology. Roughness creates eddies with a size proportional to k and a new scale separation between the roughness-scale eddies and the Kolmogorov eddies with size η, scouring the surface area. The cascading is characterised by the rate of turbulent energy dissipation $\varepsilon = u_\tau^3/\kappa k$. (b) Ensembled Kolmogorov eddies initially carry temperature Θ_i from the well-mixed region. (c) probability density function (p.d.f) of ensembled Kolmogorov eddies. The red region is the CSL. The original figure is adopted from Zhong et al. [4] and modified by the authors.

$$\frac{\partial \Theta(n,t)}{\partial t} = \alpha \frac{\partial^2 \Theta(n,t)}{\partial n^2}, \ \Theta(0,t) = 0, \ \Theta(n, \ 0) = \Theta_i, \ \Theta(n \to \infty, \ t) = \Theta_i \tag{43}$$

The solution to the diffusion equation is given by $\Theta(n, \ t) = \Theta_i \mathrm{erf}\left[n/\sqrt{4\alpha t}\right]$, which is plotted in Fig. 14 (b). Since CSL is submerged below y_i, the temperature at the edge of CSL is $\Theta_{\delta_\theta} = \Theta_i$. This approach resembles the tangent method used in the literature on Rayleigh–Bénard convection [372–374]. Applying the Fourier law of heat conduction to the diffusion equation yields, [323,324]

$$\left.\frac{\partial \Theta(n,t)}{\partial n}\right|_w \approx \frac{\Theta_i}{\delta_\theta(t)} \tag{44}$$

The solution to this equation gives $\delta_\theta(t) = \sqrt{\pi \alpha t}$. It is worth mentioning that t in this context does not represent elapsed time but rather the contact time of individual Kolmogorov eddy. The ensemble of these eddies and their respective contact times is characterised by a probability density function (PDF), denoted as PDF(t), or equivalently PDF ($\delta_\theta(t)$) [4].

In its most simplified form, the KB model for instantaneous mean flux is given by $q_w \sim \Theta_i \sqrt{\alpha/\tau}$ [324,375–377], where $\tau = t/t_\eta$. As discussed earlier, Kolmogorov eddies have random contact times, with their PDF, $P(\tau)$ satisfying the condition $\int_0^\infty P(\tau) d\tau = 1$. Here, $P(\tau)$ is a spatiotemporal characteristic of the renewal process, as evidenced by infrared surface temperature measurements [378,379]. Therefore, for each eddy, the instantaneous interfacial heat (or mass) transfer coefficient $h_i(\tau)$, can be estimated as $h_i(\tau) \sim (\alpha/\tau)^{\frac{1}{2}}$ [375,377,380]. Although this level of simplification might seem unrealistic, it is the core assumption used by Brutsaert [323]. The mean heat (or mass) transfer coefficient can then be calculated by integrating over the distribution of eddy contact times, i.e.

$$\overline{h} = \int_0^\infty h_i(\tau) P(\tau) d\tau \tag{45}$$

The average heat flux (or E in ABL literature) is obtained as $q_w = \overline{h} \ \Theta_i$.

Danckwerts [381] showed that the statistical distribution of these eddy-renewal times follows an exponential PDF with a mean contact time equal to the Kolmogorov time scale. Brutsaert [323,324] employed the exponential PDF to calculate the mean heat flux, as

$$q_w \propto \rho c_p \frac{\alpha^{1/2} u_\tau^{3/4}}{(\nu \kappa y_0)^{1/4}} \Theta_i \tag{46}$$

The 1/4 scaling $\left(\sim (\nu \kappa y_0)^{1/4}\right)$ agrees with the "universal" scaling of the liquid transfer coefficient $\left(\sim (\nu \varepsilon)^{1/4}\right)$ for sparsely soluble gases in air-water exchange [378,380,382–384]. Eq. (46) has received extensive support in the literature for various interfacial transfers of scalar quantities, including heat and mass. Specifically, it has been applied to scalar transport [323,324], evaporation from soil surfaces [385,386], mass transfer from porous media [387], and isotopic fractionation in hydrological applications [388–390].

From Eq. (46), the dimensionless interfacial temperature can be obtained as

$$\Theta_i^+ = C_R \left(k_s^+\right)^{1/4} Pr^{1/2} \tag{47}$$

where C_R is a dimensionless constant similar to α_c in CV theory, which depends on the roughness type. It is worth mentioning that in the original formula, Brutsaert [323] used the momentum roughness length y_0 rather than the roughness scale k_s. However, these two roughness scales are related by Eq. (27).

In addition to the exponential PDF employed by Brutsaert [323], various other forms of PDFs have also been explored in the literature, including power law [391,392], gamma [386,387], and inverse Gaussian [393]. The gamma distribution, in particular, has been strongly supported by experimental studies for heat and gas exchanges from soil [386,387]. Katul and Liu [363] discussed the surface renewal theory using various PDF functions. They reported that, by setting $\overline{\tau} = (\varepsilon/\nu)^{1/4}$, the interfacial heat transfer and the resulting heat flux from all PDF forms can be expressed as $\overline{h} \sim (\alpha)^{1/2} (\varepsilon/\nu)^{1/4}$ and $\overline{q_w} \sim \alpha^{1/2} u_\tau^{3/4} (\nu \kappa y_0)^{1/4} \Theta_i$, respectively, confirming Brutsaert's results.

Katul and Liu [363] concluded that the shape of the PDF may have a marginal impact on the bulk heat transfer coefficient and proposed a new approach that links the eddy turnover velocity with the vertical velocity structure function, which is particularly effective for cases where the molecular Prandtl (or Schmidt) number is equal to or greater than one.

7.3. Further discussion on heat transfer models

Both the CV hypothesis and KB theory were developed based on the assumption of a fully rough flow regime. In this regime, the flow is independent of viscous friction, and the Reynolds number is sufficiently high to sustain a complete logarithmic layer. However, these models are not generally applicable to transitionally rough regimes, where the flow does not fully satisfy these conditions. Moreover, when $Pr \ll 1.0$, as in the case of liquid metals, the thermal boundary layer may extend beyond the interfacial height, y_i. In such cases, molecular conduction becomes significant within the logarithmic profile, challenging the assumptions underlying the interfacial concept [231,232]. Consequently, these models are not generally applicable to flows with $Pr \ll 1.0$.

The CV model operates under the assumption of the Reynolds analogy, which presumes negligible molecular diffusion, with heat and mass

transport primarily facilitated by fluid motion. However, this assumption breaks down near the surface for two main reasons. Firstly, at the surface, the flow velocity drops to zero, while the average temperature (or concentration) remains finite, leading to a discrepancy in transport mechanisms. Secondly, in the fully rough regime, a thin CSL conforms closely to the surface roughness topography, a phenomenon recently confirmed by DNS studies [4,61,66,68]. MacDonald et al. [66] suggested that the temperature field near a rough wall behaves as though the surface were effectively smooth but with an increased surface area. This modification influences the heat transfer characteristics, differentiating it from a truly smooth wall.

Establishing universal length and velocity scales is crucial for developing a unified theory of roughness. The original CV hypothesis by Owen & Thomson [191] was formulated under the assumption that u_τ and k_s represent the characteristic velocity and length scales of the sublayer, respectively. Dipprey & Sabersky [297,330] used the same characteristic length scale but proposed an alternative characteristic velocity u_g, defined as the velocity at the roughness peaks. However, no explanation was provided for the choice of these specific length scales. Since $u_g = Au_\tau$, it appears that u_τ could serve as a universal velocity scale within the roughness layer, which also used in KB theory. The characteristic length in the KB theory is y_0, which is proportional to k_s. According to wall similarity hypothesis, roughness effects remain within the roughness sublayer, extending from the wall up to $3 - 5k_s$ [247]; therefore, the assumption of k_s as the length scale seems reasonable. In conclusion, rough-wall flow within the roughness layer can be scaled with the length scale k_s and the velocity scale u_τ, used for scaling St_k^{-1}. Therefore, St_k^{-1} seems to be a better option than g_k for roughness heat transfer scaling. Several studies [63,68,297,318] have found an anlytical relationship between St_k^{-1} and RA factor, as stated Eq. (38), which can be used to obtain St_k^{-1} in experimental and CFD studies.

The phenomenological model proposed by Yaglom & Kader [346], as expressed in Eq. (42), is noteworthy because it does not require detailed knowledge of the roughness sublayer. This model is also valid for high Prandtl number flows, as demonstrated by the experimental studies of Park & Chung [322], which demonstrated similar scaling for $Pr = 2014$. However, it is important to note that the assumptions underlying this model are not generally applicable for $Pr \ll 1.0$.

Most studies on rough-wall heat transfer have concentrated on air ($Pr \approx 0.7$) and, to a lesser extent, water ($Pr \approx 2 - 6$). As a result, there is a significant gap in the testing and validation of these theories across a broad spectrum of (k_s^+, Pr) values, particularly for flows with low and high Prandtl numbers. This lack of comprehensive testing leaves an open question regarding the applicability of existing models across different flow regimes and Prandtl numbers.

The discussion in the literature focuses on determining the appropriate values for the exponents and the coefficient of heat transfer enhancement $\approx \alpha_c \left(k_s^+\right)^p Pr^m$. According to the CV hypothesis, typical values are $p = 1/2$ and $m = 2/3$. Conversely, the KB theory suggests $p = 1/4$ and $m = 1/2$. The question of universal scaling for heat transfer remains an ongoing debate in the literature.

Zhong et al. [4] conducted a DNS study on convective flow over rough walls with $Pr = 0.5$, 1.0, and 2.0 within $14.85 < k_s^+ < 30$. Their findings indicated that the mean heat transfer aligns with the KB theory, where they determined $p = 1/4$ and expressed the interfacial temperature as [4]

$$\Theta_i^+ = 3.7\left(k_s^+\right)^{1/4} Pr^{1/2} + 2.6 \tag{48}$$

They also observed that in regions of locally attached high-shear flow—such as around the roughness peaks—the CV hypothesis was confirmed. In contrast, no clear correlation between heat and momentum transfer was observed in the reversed flow region behind the roughness elements. These findings led Zhong et al. [4] to conclude that the overall understanding of heat transfer in fully rough conditions cannot be fully explained by a single mechanism or phenomenon.

Almost all literature studies agree that the coefficient α_c depends on the roughness type. Table 2 highlights the wide range of α_c values proposed for different roughness types, indicating that a universal α_c must account for the roughness topological parameters discussed in Section 5. Flow impingement on windward face of roughness and flow separation on the wake region behind roughness impact heat transfer. Heat transfer tends to increase on the windward slopes due to the impingement of unmixed fluid with high-temperature gradients, while it decreases on the leeward slopes due to the presence of well-mixed fluid recirculation with lower temperature gradients [61]. Therefore, it is expected that α_c is influenced by spatial and spectral topologies, with roughness density having a particularly significant impact on heat transfer. Roughness frontal density (λ_f) is recognised as a key factor controlling impingement and recirculation [285]. Moreover, in the fully rough regime, heat transfer generally peaks within the range of $0.1 < \lambda_f < 0.3$ [64,168]. Unfortunately, there is limited data on the impact of these roughness topologies on heat transfer.

Rowin et al. [68] suggested that local flow and heat transfer can be described by a combination of "sheltered" and "exposed" behaviours. Here, "exposed" refers to the flow regions influenced by a favourable pressure gradient (high shear), while "sheltered" pertains to regions characterised by an adverse pressure gradient (flow reversal). They decomposed the wetted area (A_w) into the exposed and sheltered areas ($A_w = A_e + A_s$), and obtained

$$St_k = \frac{A_w}{A_t}\left[\left(\frac{A_e}{A_w}\right)St_{k,e} + \left(\frac{A_s}{A_w}\right)St_{k,s}\right] \tag{49}$$

where $St_{k,e}$ and $St_{k,s}$ are the exposed and sheltered roughness Stanton numbers, respectively, which are defined based on the u_k and Θ_k which are the velocity and temperature at the level of roughness peaks, respectively. The ratio A_e/A_t is a function of λ_f. Rowin et al. [68] used the scaling proposed by Zhong et al. [4] for high shear flow over exposed elements and suggested

$$St_{k,e} = 0.5\left(\lambda_f\right)^{1/2}\left(k^+\right)^{-1/2} Pr^{-2/3} \tag{50}$$

$$St_{k,s} = 0.012 Pr^{-0.45} \tag{51}$$

where $0.5 \le Pr \le 2$ and k^+ is based on the roughness height. This scaling also performs well in both fully rough as well as transitionally rough regimes where $k^+ \approx O(10)$. Rowin et al. [68] observed that $St_{k,s}$ is independent of k^+ and approaches a constant value in the fully rough regime. They also noted that the exposed contribution is dominant, and their model was deemed adequate by assuming $St_{k,s} = 0$.

Recent experimental study by Kadivar et al. [394] demonstrated that for roughness with high thermal conductivity, such as additive-manufactured roughness made of aluminium, the Nusselt number scales with $Pr^{0.8}$ as compared to smooth cooling channels which generally scale with $Pr^{0.4}$. They suggests that the thermal conductivity of roughness can dominate the energy transfer from the solid wall to the flow near roughness.

8. Rough-wall temperature profile

The primary distinction between heat transfer and momentum transfer lies in the absence of an analogous relationship for heat transfer, similar to the direct pressure-drag analogy in momentum transfer. In the fully rough regime, unlike the wall shear stress (τ_w), the wall heat flux (q_w) does not scale with the inertial flow governed by the roughness elements (y/k_s). Instead, q_w remains sensitive to the Prandtl number, alongside the characteristics of the flow itself. This difference underscores the separate impacts of roughness and molecular properties on convective heat transfer under fully rough conditions.

Leonardi et al. [395] conducted a DNS study of channel flow with

two-dimensional, spanwise-aligned square bars on the bottom wall and a smooth surface on the top wall. They observed different temperature profiles on each wall, with slightly varying logarithmic slopes depending on the aspect ratios of the bars. Chan et al. [162] suggested that the asymmetric conditions between the bottom and top wall might influence the perception of outer-layer similarity in the flow, possibly explaining the different slopes observed in the temperature profiles. Pirozzoli et al. [190] observed significant wakes in the outer layer of temperature profiles in channels with differing prescribed temperatures between the bottom and top walls. This wake formation can obscure the identification of the logarithmic region, particularly at the relatively low friction Reynolds numbers investigated in the roughness cases studied by Leonardi et al. [395]. MacDonald et al. [66] also conducted a DNS study and confirmed the logarithmic behaviour of the temperature profile.

As an analogy to the velocity shift observed in turbulent flows, the logarithmic temperature profile over a rough wall can be derived by shifting the smooth wall profile. Roughness usually increases heat transfer relative to a smooth wall, resulting in a downward shift of the logarithmic temperature profile [217,239,293,396].

$$\Theta_R^+ = \Theta_S^+ - \Delta\Theta^+\left(k_s^+, Pr\right), \tag{52}$$

where $\Delta\Theta^+$ is the temperature shift.

In the literature, the logarithmic temperature profile has been established using various dimensionless coordinates, including viscous scaling, y^+, [63,92,94,194], momentum length scaling, y/y_0, thermal length, $y/y_{0,\theta}$, [397], interfacial sublayer, y/y_i, [191,323,346,398], and sand-grain roughness scaling, y/k_s, [191,297,319,346]. Correspondingly, alternative formulations of the temperature profile have been proposed [194,297,361,396], expressed as

$$\Theta_R^+ = \frac{1}{\kappa_\theta}\ln(y-d)^+ + \beta(Pr) - \Delta\Theta^+\left(k_s^+, Pr\right), \tag{53}$$

$$\Theta_R^+ = \frac{1}{\kappa_\theta}\ln\left(\frac{y-d}{y_0}\right) + St_k^{-1}\left(k_s^+, Pr\right), \tag{54}$$

$$\Theta_R^+ = \frac{1}{\kappa_\theta}\ln\left(\frac{y-d}{y_{0,\theta}}\right), \tag{55}$$

$$\Theta_R^+ = \frac{1}{\kappa_\theta}\ln\left(\frac{y-d}{y_i}\right) + \Theta_i^+\left(k_s^+, Pr\right), \tag{56}$$

$$\Theta_R^+ = \frac{1}{\kappa_\theta}\ln\left(\frac{y-d}{k_s}\right) + g_k\left(k_s^+, Pr\right). \tag{57}$$

Similar to the challenges of obtaining ΔU^+ in terms of k_s^+ for momentum transfer, the heat transfer problem is reduced to finding expressions for $\Delta\Theta^+$, St_k^{-1}, Θ_i^+ or g_k in terms of k_s^+ and Pr.

Relationships can be established between various roughness scales for different y scaling. Earlier, a relationship between y_0 and k_s was established in Eq. (27). The thermal length scale $y_{0,\theta}$ is the roughness length for heat flows [3], commonly utilised in meteorology and ABL flows. By comparing Eq. (54) and Eq. (55), a relation between $y_{0,\theta}$ and y_0 can be obtained as $y_0/y_{0,\theta} = \exp\left(\kappa_\theta St_k^{-1}\right)$. This relation combined with Eq. (27) yields

$$\frac{k_s}{y_{0,\theta}} = \exp\left(\kappa_\theta St_k^{-1} + \kappa B\right), \tag{58}$$

For the fully rough regime, $k_s/y_{0,\theta} \approx 30\exp\left(\kappa_\theta St_k^{-1}\right)$. Although y_0 and $y_{0,\theta}$ are related, they differ significantly because momentum transfer and heat transfer exhibit distinct behaviours in the roughness layer. This discrepancy arises from the fact that, near the wall, heat transfer is influenced primarily by molecular diffusion, whereas momentum transfer is additionally governed by pressure forces [3]. A significant body of research in meteorology and ABL flows has focused on investigating the relationship between y_0 and $y_{0,\theta}$, or alternatively determining $\kappa_\theta St_k^{-1}$ (see reviews by Massman [399] and Verhoef et al. [400]). Various methods have been proposed to express $\kappa_\theta St_k^{-1}$ in terms of known external flow parameters and roughness (canopy) characteristics. These methods include simple relations based on dimensional analysis [3,335,401,402], complex models of heat transport in the roughness layer [397,399,402–405], and data science approaches using machine learning algorithms trained on eddy covariance data [406].

By combining Eq. (54) and Eq. (56), it can be shown that

$$\Theta_i^+ = \frac{1}{\kappa_\theta}\ln\left(\frac{y_i}{y_0}\right) + St_k^{-1}. \tag{59}$$

Unlike y_0 and $y_{0,\theta}$, the interfacial length y_i does not have any fixed relation with y_0 or k_s. Some researchers [191,361] suggested $y_i = k_s$ as a starting point.

A relation for g_k can be obtained by combining Eq. (53), (57), and (27), written as

$$g_k = St_k^{-1} + Pr_t B; \tag{60}$$

For fully rough regimes, $g_k = St_k^{-1} + 8.5Pr_t$. Originally, g_k was introduced by Dipprey & Sabersky [297], who suggested $g_k = St_k^{-1} + A$, and they assume $A = 8.48$ to establish a limiting case. Interestingly, by assuming $Pr_t \approx 1.0$, Eq. (60) simplifies to the relation proposed by Dipprey & Sabersky. By considering $y_i = k_s$ (suggested by various researchers [61,66,191,361], and combining Eq. (59) and (60), it follows that $\Theta_i^+ = g_k$.

8.1. The impact of wall origin

The wall origin, discussed in Section 2.3, is critical in cases where roughness occupies a significant portion of the boundary layer as well as pipe and channel flows that the boundary layer growth is limited by the channel dimension.

Fig. 15 compares the temperature profiles from DNS studies by Peeters and Sandham [61] and Kuwata [63]. Both studies investigated turbulent heat transfer in channels with irregular roughness under a uniform wall temperature boundary condition and $Pr = 1$. Both studies used mean peak-to-valley heigh (S_t) as roughness scale which is approximately equal to k_s for roughness with near Gaussian profile [186,256,258]. Both figures demonstrate a downward shift in mean temperature profiles with increasing k_t^+. However, there are distinct differences in the near-wall temperature profiles, likely due to differences in boundary conditions.

Peeters and Sandham [61] data (Fig. 15 (a)) shows an upward shift of the temperature profile with increasing k_t^+ for $y^+ \lesssim 10$. They interpreted this behaviour as an augmentation of heat transfer towards $y^+ = 0$. On the other hand, Kuwata [63] (Fig. 15 (b)) shows that the entire temperature profile experienced a downward shift, attributed to enhanced heat transfer by roughness. A similar behaviour was observed by Kuwata [62] and Garg et al. [92–94]. In Fig. 15 (a), the wall origin is the mean value of roughness; however, in Fig. 15 (b) the temperature profile was plotted versus the normal distance y_e (see section 2.3).

Zhong et al. [4] conducted a DNS study with $Re_\tau = 395$ to 2000 and $Pr = 0.5$, 1, and 2, using a mixed boundary condition defined as $T_w = T + \theta_w = T_{ref} + dT/dx$, where T_{ref} is the reference temperature. Fig. 16 illustrates the temperature profiles by Zhong et al. [4] for different Prandtl numbers. The contrasting shades of lines correspond to increasing k_s^+, while the dashed lines represent the smooth wall profile at $Re_\tau = 2000$. Macdonald et al. [66] also showed similar temperature below the roughness crests constantly increases with k_s^+ in the fully rough regime. Fig. 16 highlight a well-mixed region below the logarithmic profile.

Zhong et al. [4] showed that, for $14.85 < k_s^+ < 300$, the wall origin is located above the mean roughness height and approaches the roughness peaks as k_s^+ increases. This suggests that, for higher k_s^+ values, the wall

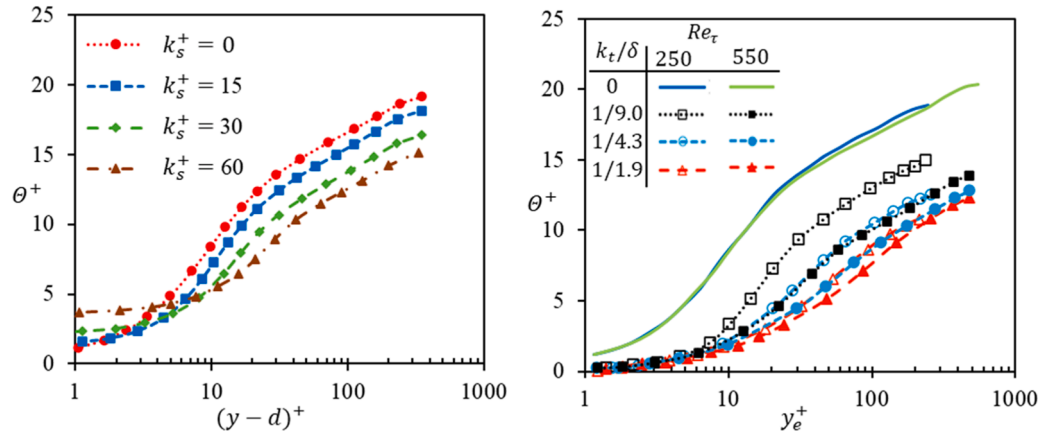

Fig. 15. Temperature profiles from the DNS of (a) Peeters and Sandham [61] and (b) Kuwata [62] at $Pr = 1$.

origin might be located above the roughness peaks. They also reported that the uncertainty associated with the location of the wall origin does not substantially influence the evaluation of ΔU^+ and $\Delta\Theta^+$, and the temperature and velocity wall-origin roughly collapse to the same values.

MacDonald et al. [60] and Rowin et al. [68] used the same 3D sinusoidal roughness at $Pr = 0.71$ and used different methods to determine the virtual origin (assuming $d_\theta = d$ in both studies), MacDonald et al. [60] reported $\Delta\Theta^+ \approx 4.4$, while Rowin et al. [68] reported $\Delta\Theta^+ \approx 3.8$, resulting in a difference of 13.6%, which is not negligible. This discrepancy likely stems from methodological choices rather than physical differences between the studies [57].

8.2. Evaluating $\Delta\Theta^+$

Various methods can be employed to evaluate the temperature shift $\Delta\Theta^+$. One approach in using the difference between the outer layer temperature of smooth and rough wall flows at a matched friction Reynolds number. One approach involves using the temperature difference between the outer layer of smooth and rough wall flows at a matched friction Reynolds number. A more conservative method, however, is to compare the mean streamwise temperature difference between the smooth and rough walls at the same friction Reynolds number [92,94], i.e.

$$\Delta\Theta^+ = \Theta^+_{a,S} - \Theta^+_{a,R}. \tag{61}$$

The relation between Θ^+_m and Θ^+_a can be expressed by [66]

$$\Theta^+_m - \Theta^+_a = \frac{Pr_t}{\kappa^2 U^+_b}. \tag{62}$$

A relation for the temperature shift as a function of St and C_f can be derived by substituting Eq. (8), (9), and (62) into Eq. (61), which results in

$$\Delta\Theta^+ = \sqrt{\frac{C_{f,S}}{2}}\left(\frac{1}{St_S} - \frac{Pr_t}{\kappa^2}\right) - \sqrt{\frac{C_{f,R}}{2}}\left(\frac{1}{St_R} - \frac{Pr_t}{\kappa^2}\right), \tag{63}$$

Using the inner-scaled bulk mean velocity and arithmetic mean temperature, Kuwata [63], derived

$$\Delta\Theta^+ = Pr_t(\Delta U^+ - C) + \beta(Pr) + \frac{Pr_t}{\kappa^2}\sqrt{\frac{C_f}{2}} - St_k^{-1} \tag{64}$$

Rowin et al. [68] analytically derived a slightly different relation, written as

$$\Delta\Theta^+ = \underbrace{Pr_t(\Delta U^+ - C) + \beta(Pr)}_{Term\ 1} - \underbrace{\sqrt{\frac{2}{C_f}}\left(\frac{C_f}{2St} - Pr_t\right)}_{Term\ 2} \tag{65}$$

where C is the smooth-wall intercept of the velocity profile. This expression distinguishes between two types of contributions: Term 1 relies solely on inner-scale quantities that remain constant regardless of the bulk (outer-scale) Re, while Term 2 is concerned only with the bulk quantities (C_f and St), which directly depends on the bulk Re. The Reynolds number does not appear explicitly, suggesting that $\Delta\Theta^+$ does not scale with Re. Revisiting Eq. (38), it can be shown that Term 2 $= St_k^{-1}$, and by utilising Eq. (24), we derive as general formula for $\Delta\Theta^+$, written as

$$\Delta\Theta^+ = \frac{1}{\kappa_\theta}\ln k_s^+ + [Pr_t B + \beta(Pr)] - St_k^{-1} \tag{66}$$

This expression has a similar structure to the empirical correlation of Kays and Crawford [194],

$$\Delta\Theta^+ = \frac{1}{\kappa_\theta}\ln(k_s^+) + \left[\beta(Pr) - 3.484\frac{Pr_t}{\kappa}\right] - \left[1.25(k_s^+)^{0.20}Pr^{0.44}\right]. \tag{67}$$

8.3. The behaviour of $\Delta\Theta^+$

While ΔU^+ is solely a function of k_s^+, $\Delta\Theta^+$ varies with both k_s^+ and Pr. In y^+ coordinates, the log intercept $\beta(Pr)$ is determined such that $\Delta\Theta^+ = 0$ for smooth walls. A positive $\Delta\Theta^+$ represents heat transfer enhancement due to roughness. When both k_s^+ and $Prk_s^+ \ll 1$, the enhancement in heat transfer due to roughness is negligible [396].

The number of studies on $\Delta\Theta^+$ significantly lags those on ΔU^+. A substantial body of literature has focused on studying ΔU^+ and characterising it across various roughness topologies and flow regimes. Reviews of these studies can be found in Bradshaw [407], Flack and Schultz [117], Andersson et al. [408], Kadivar et al. [52] and Chung et al. [51]. However, research on the behaviour of $\Delta\Theta^+$ is limited, with current studies largely limited to $Pr \approx 0.7 - 1.0$.

The measured or calculated ΔU^+ values for various roughness types collapse onto a single universal asymptote using the concept of k_s^+ as the reference roughness scale (see Chung et al. [51] and Kadivar et al. [52]). Fig. 17 (a) compares selected $\Delta\Theta^+$ values from different DNS and LES studies, plotted against k_s^+ for different Pr, symbols are listed in Table 3. An asymptotic behaviour is observed for most of the data in the fully rough regime, where $\Delta\Theta^+$ tends to approach a constant value (which is dependent on Pr) as k_s^+ increases. This asymptotic behaviour has been suggested by MacDonald et al. [66] and Hantsis & Piomelli [57]. However, due to insufficient data, it is uncertain whether this behaviour

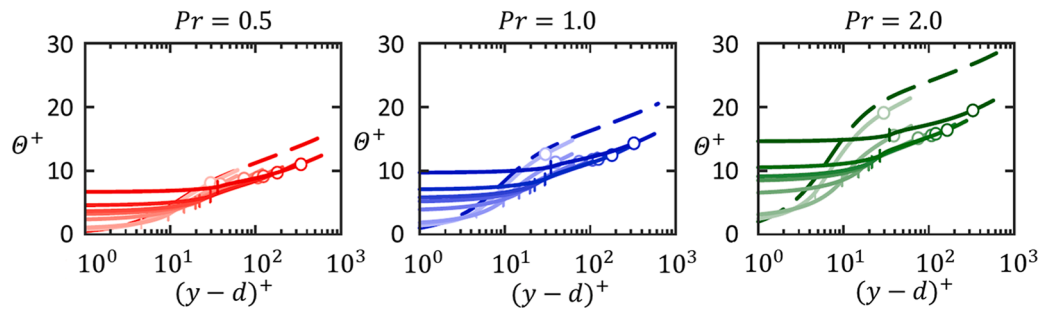

Fig. 16. Temperature profiles by Zhong et al. [4] for various Prandtl numbers using mixed boundary conditions. Darker lines indicate increasing k_s^+ and the dashed lines represent the temperature profile of a smooth wall at $Re_\tau = 2000$ and the circle markers indicate estimates for the start of the logarithmic region.

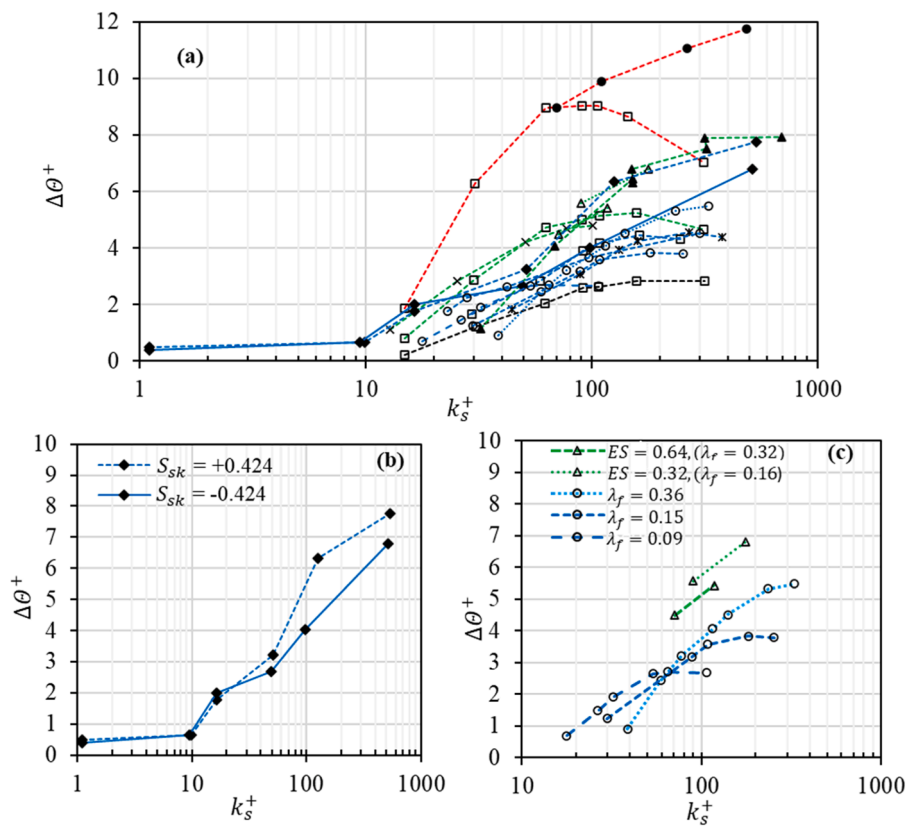

Fig. 17. Behaviour of $\Delta\Theta^+$ with (a) different roughness types and Prandtl numbers, (b) with skewness, (c) with roughness density. The symbols are listed in Table 3. The black, blue, green, and red couleurs are associated with $Pr = 0.5$, 0.7, 1.0, and 2.0, respectively.

persists at very high k_s^+ values. The data from Zhong et al. [4] suggests that $\Delta\Theta^+$ may reach a peak at a specific k_s^+, after which it decreases with further increases in k_s^+. The potential for a decreasing trend in $\Delta\Theta^+$ at higher k_s^+ cannot be conclusively supported by the current data available in the literature. However, if such a decrease were to continue, it would imply that $\Delta\Theta^+$ could eventually become negative, suggest in a reduction in heat transfer relative to a smooth wall.

Neither the asymptotic behavior nor the decreasing trend is predicted by traditional models, such as the "CV hypothesis" or "KB Theory." The correlations presented in Table 2 also do not allow for asymptotic behaviour or a decreasing trend with k_s^+. This limitation arises because these models are typically developed by fitting limited data from specific roughness types and a restricted range of k_s^+ and Pr; therefore, their general applicability across different roughness types cannot be guaranteed (see, e.g. Refs [116,191,297,319,334,351]). Phenomenological models (e.g. Yaglom & Kader [239], Gioia & Chakraborty [409], Peeters [410], Owen & Thomson [191], and Brutsaert [3]), which, unlike empirical models, incorporate physical assumptions, may provide a more robust basis for scrutiny.

The trends of $\Delta\Theta^+$ in Fig. 17 indicates that heat transfer augments by increasing Pr. It can also be argued that the dependency of $\Delta\Theta^+$ on Pr is greater than that on k_s^+ and roughness topology, such as λ_f and S_{sk}. Fig. 17 (b) and (c) shows the dependency of $\Delta\Theta^+$ to roughness skewness and density, respectively, highlighting the importance of accounting for these parameters in evaluating $\Delta\Theta^+$, and thus heat transfer. As shown in Fig. 17 (b), positive skewness increases heat transfer more than that of negative skewness. Additionally, the effect of skewness seems to be marginal in the transitionally rough regime. Fig. 17 (c) demonstrates the variation of $\Delta\Theta^+$ with frontal solidity λ_f behaves differently between the transitionally rough and fully rough regimes. In the transitionally rough regime, higher $\Delta\Theta^+$ values correspond to lower λ_f. Conversely, in the fully rough regime, this trend is reversed, with higher λ_f leading to increased $\Delta\Theta^+$.

9. Wall similarity hypothesis

Wall similarity is derived from Townsend's hypothesis [196] and it is an important concept in turbulent flow, widely used to describe turbulent flow over both smooth and rough surfaces. Townsend proposed that near-wall turbulence consists of two regions: the "active" region, responsible for momentum transfer, and the "inactive" region, which does not influence this transfer. Active turbulence is shear-driven and depends on local flow conditions, while inactive turbulence is generated far from the surface and scales with the outer layer's parameters.

The wall similarity hypothesis states that, in turbulent flow over both rough and smooth surfaces, the turbulent motions outside the roughness (or viscous) sublayer become independent of the surface roughness and viscosity at high Reynolds numbers. This suggests that the normalised turbulent stress, relative to wall shear stress, remains consistent beyond the roughness sublayer and collapses to the smooth-wall data. Therefore, roughness effects are limited to the roughness layer and the outer layer remains unaffected by roughness. The wall similarity hypothesis has been supported by numerous DNS and experimental studies on various types of roughness [52,244].

Some researchers have reported exceptions to the wall similarity hypothesis, suggesting that the outer layer may be sensitive to surface conditions [93,240,244,411–413]. Flack et al. [244] and Garg et al. [93] highlighted exceptions to the wall similarity hypothesis and attributed these discrepancies to the use of "strong" roughness in certain studies. In flows with strong roughness, a significant portion of the log-law region is protruded by roughness elements. Flack et al. [244] highlighted exceptions to the wall similarity hypothesis, attributing them to "strong" roughness, where the roughness height is a significant part of the inner layer. In such cases, the roughness effects can extend into the outer layer, disrupting the flow. Jiménez [50] supported this, stating that rough-wall flows with $\delta/k \leq 40$ might affect the outer layer. Flack et al. [244] suggested a similar criterion, $\delta/k_s < 40$. They also showed that Reynolds stresses align well in most of the boundary layer, with turbulence variations confined to $y < 5k_s$.

While wall similarity for momentum boundary layer is detailly discussed in the literature (see, e.g. Jiménez [50], Flack et al. [244], and Kadivar et al. [52]), the wall similarity for the thermal boundary layer has received less attention. According to wall similarity, the defect temperature profile should only be a function of the normal wall distance (y), represented by

$$\Theta_e^+ - \Theta^+ = f\left(\frac{y}{\delta}\right), \tag{68}$$

where Θ_e^+ is the temperature at the edge of the boundary layer or the centere of the channel.

Fig. 18 illustrates the defect temperature profile using DNS data from Peeters & Sandham [61] and Kuwata [63]. It demonstrates significant discrepancies in the roughness layer, but for $y/\delta \geq 0.4$, the profiles show a closer match. The Peeters & Sandham [61] data shows better agreement in the temperature profiles starting closer to the wall for $y/\delta \geq 0.06$. A closer view of the defect temperature profiles in the inset of Fig. 18 shows that, even near the channel centre, wall similarity is not perfectly preserved. This may be due to the strong roughness used in these studies, with $\delta/k \approx 7.1$ used by Kuwata [63] and $\delta/k \approx 6$ employed by Peeters & Sandham [61]. Further studies are required to gain a better understanding of wall similarity for the thermal boundary layer.

10. Thermal efficiency of roughness

Previous studies [188,297,346,414] have shown that roughness increases heat transfer coefficient (St); however, it is accompanied by a significant increase in friction drag (C_f). This indicates that the RA factor for some rough-wall flows can be smaller than that for smooth walls.

Fig. 19 (a) compares St_R/St_S and $C_{f,R}/C_{f,S}$. In most cases, the friction drag increases faster than the Stanton number, implying $RA_R < 1$. However, the data of Kuwata [208] shows an opposite behaviour. It is worth mentioning that Kuwata [208] utilised high-aspect-ratio streamwise ribs, whose behaviour differs significantly from that of distributed roughness due to the Kelvin-Helmoltz (KH) instability generated rollers that suppress sweeps and ejections, leading to lower Reynolds stress. Rouhi et al. [415] also demonstrated that KH rollers play a favourable role in locally breaking the Reynolds analogy by creating negative wall shear stress patches in the riblet grooves, a phenomenon without a counterpart in wall heat flux. In contrast, secondary flows are responsible for unfavourable local deviations from the Reynolds analogy, particularly near ribblet crests, as observed by Modesti et al. [416]. Taken together, these studies highlight the interplay between KH rollers and secondary flows in determining the global performance of the Reynolds analogy, with KH rollers enhancing and secondary flows diminishing heat-transfer efficiency. $RA_R > 1$ is achievable only within a specific Re range; beyond a certain rib spacing (p^+). This holds for different longitudinal ribs and riblets, with a p^+ range for $RA_R > 1$ varying by geometry [208,415,416].

Bons [126] proposed considering an RA efficiency defined as the ratio of the RA factor of the rough wall to that of a smooth wall, i.e., RA_R/RA_S. An RA efficiency smaller than 1.0 indicates that the impact of roughness on momentum transfer is greater than that on heat transfer, implying a more significant increase in friction due to roughness compared to enhanced heat transfer. Based on Townsend's hypothesis for zero pressure gradient boundary layer, Aupoix [107] derived

$$\frac{RA_R}{RA_S} = \frac{1 - \Delta U^+ \sqrt{\frac{C_{f,R}}{2}}}{1 - RA_s \Delta\Theta^+ \sqrt{\frac{C_{f,S}}{2}}} \tag{69}$$

The impact of roughness is considered through ΔU^+ and $\Delta\Theta^+$, which are only a function of k_s^+ (at constant Prandtl number). Although $C_{f,S}$ depends on the Reynolds number, in the fully rough regime, its dependency is not significant. Therefore, based on Eq. (69), at a constant Prandtl number, the RA efficiency can be solely characterised by k_s^+. Based on this rationale, Forooghi et al. [55] conducted a DNS study on various roughness types, including realistic roughness geometries, and suggested

$$\frac{RA_R}{RA_S} = 0.55 + 0.45\exp\left(-\frac{k_s^+}{130}\right) \tag{70}$$

This relation is valid within a 10% error interval which is also verified by Kuwata [63].

Peeters and Sandham [61] conducted a DNS study on convective heat transfer over irregular roughness and compared their results with the relation proposed by Aupoix [107] and Forooghi et al. [55]. They demonstrated that the relation by Forooghi et al. [55] agrees with their data with a discrepancy of up to 2.35%. They noted that the Aupoix [107] relation can provide excellent results within approximately ±1.4% when their DNS results are used for $\Delta\Theta^+$. This indicates the importance of accurate knowledge of $\Delta\Theta^+$.

Fig. 19 (b) shows the variation of RA efficiency (RA_R/RA_S) against k_s^+ for various roughness types. It shows that RA_R/RA_S decreases with an increase in k_s^+, eventually reaching an asymptotic value at sufficiently high k_s^+ values. Eq. (70) predicts all the roughness morphologies considered by Forooghi et al. [417] within a 10% error, as well as the results of Jelly et al. [421], Peeters and Sandham [61], Kuwata and Kawaguchi [167], and Garg et al. [92,94]. A figure similar to Fig. 19 (b) was discussed in a recent review by Hantsis and Piomelli [57]. However, there is a notable difference between the findings of the present study and those reported by Hantsis and Piomelli [57] concerning the data from Garg et al. [92]. In their review, Hantsis and Piomelli [57] presents the RA_R/RA_S values from Garg et al. [92] as significantly lower than other literature data and stated this discrepancy is attributed to the unique characteristics of additive-manufactured roughness. In contrast,

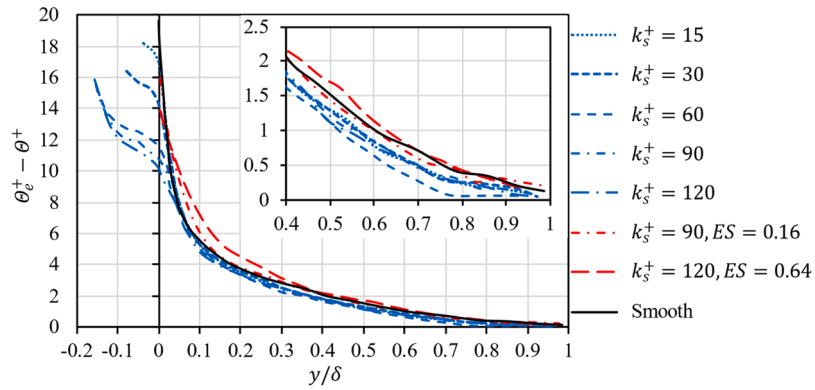

Fig. 18. Comparison of defect temperature profiles from rough-wall DNS data of Peeters & Sandham [61] (blue lines) and Kuwata [63] (red lines) with smooth-wall DNS of Kuwata [63] (black solid lines).

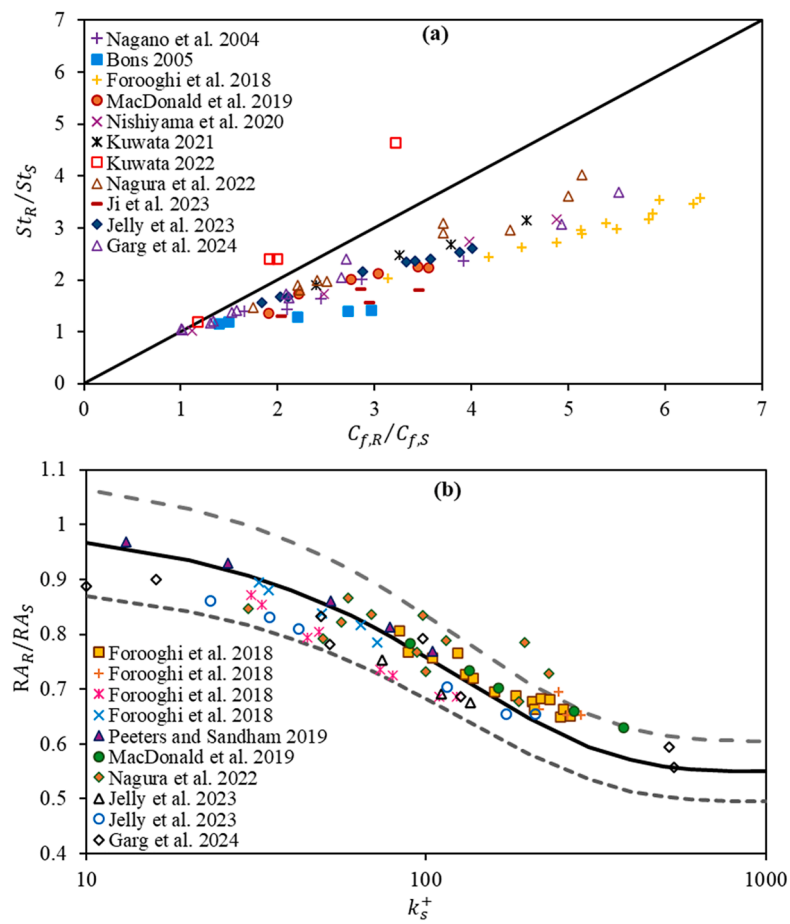

Fig. 19. (a) Comparison of St_R/St_S versus $C_{f,R}/C_{f,S}$ with the black solid line indicates $St_R/St_S = C_{f,R}/C_{f,S}$. (b) Variation of thermal efficiency of roughness, RA_R/RA_S, with the non-dimensional equivalent sand-grain roughness, k_s^+. The solid black line represents Eq. (70) with grey dashed lines indicating a 10% error interval. The data adopted from irregular 3D printed roughness by Garg et al. [92,94], longitudinal ribs by Kuwata [208], turbine blades roughness by Bons [126], irregular roughness by Forooghi et al. [417], sinusoidal roughness from [66], anisotropic porous media by Nishiyama et al. [418], irregular roughness by Kuwata [63], V-shaped ribs by Ji et al. [419], irregular roughness by Kuwata & Kawaguchi [167], irregular roughness by Peeters & Sandham [61], sinusoidal roughness from [66], sinusoidal roughness by Nagura et al. [420], irregular roughness by Jelly et al. [421].

the current study (Fig. 19 (b)) utilises the original data from Garg et al. [92,94], demonstrating that it falls within the expected range of literature values. There may have been an underestimation of the RA_R/RA_S values by Hantsis and Piomelli [57] when calculating from the available data in Garg et al. [92].

11. The behaviour of Turbulent Prandtl number and turbulent viscosity

The turbulent Prandtl number, Pr_t, is an important factor that is associated with the stability in the atmospheric stable boundary layer (SBL), defined by $Pr_t = \nu_t/\alpha_t$, where ν_t is the turbulent viscosity and α_t is the turbulent thermal diffusivity. The turbulent Prandtl number describes the difference in turbulent transfer between momentum and

sensible heat. As mentioned earlier, turbulent momentum transfer is more efficient than turbulent heat transfer when $Pr_t > 1$ and vice versa. Pr_t also demonstrates the difference between the stability profile functions of momentum and sensible heat when $Pr_t \neq 1$. It is an important characteristic of momentum and heat turbulent mixing for calibrating most empirical correlations as well as eddy diffusivity turbulence models and other applications [92,94]. Pr_t can be used to determine the eddy-diffusivity for heat directly from eddy-diffusivity of momentum, to identify the coefficients in more complex models [422,423], or to calculate dispersive fluxes [424]. A significant body of literature has been devoted to determining Pr_t in various flows, with particular interest in canonical wall-bounded [425] and ABL flows [426–429].

Kays [221] discussed the usefulness, effectiveness, and limitations of using Pr_t. He reviewed different methods for calculating Pr_t by collecting a wide range of sources (analytical, experimental and numerical) for canonical flows such as boundary layers, pipes, and channels. For common engineering applications (e.g., heat exchanges), Pr_t is estimated via empirical or semiempirical relations. A core conclusion by Kays [221] is that Pr_t is effectively constant in the logarithmic region near the wall and its value is primarily a function of Re and Pr. Generally, Pr_t is obtained in thin shear-layer flows, from the only non-zero Reynolds shear stress or zero scalar flux and velocity or scalar gradient, a more general form is obtained through a contraction with the strain rate or scalar gradient [430]:

$$Pr_t = \frac{\nu_t}{\alpha_t}, \nu_t = -\frac{\langle \overline{u'_i u'_j} \rangle \left[\overline{S}_{ij}\right]}{(2\langle \overline{S}_{kl} \rangle \langle \overline{S}_{kl} \rangle)}, \alpha_t = -\frac{\langle \overline{u'_i \theta'} \rangle \left\langle \frac{\partial \overline{\theta}}{\partial x_i} \right\rangle}{\left(\left\langle \frac{\partial \overline{\theta}}{\partial x_m} \right\rangle \right)^2}, \tag{71}$$

where $\langle \overline{S}_{ij} \rangle = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$ is the strain-rate tensor. When Pr is near unity, Pr_t approaches a value of about 0.85 – 0.9 and can be assumed constant for high enough Re [190,221,226,227]. However, the same studies indicate that, when approaching the wake region of a boundary layer, the channel centreline, or the pipe axis, Pr_t decreases to values around 0.5. In the context of CFD, many models assume a constant value for Pr_t, even though this is only justified for the logarithmic region. All terms constituting Pr_t are influenced by roughness within the roughness sublayer, consistent with Townsend's outer-layer similarity hypothesis [196]. Consequently, various quantities converge onto the smooth-wall curves and the differences in Pr_t between smooth and rough wall cases are confined to the roughness sublayer. Relatively little attention has been given to Pr_t in the context of rough-wall flows: riblets [64,431], bed roughness [432], regular roughness [66,433,434], irregular artificial roughness [61–63], and irregular 3D printed roughness [94].

In the presence of roughness, Pr_t exhibits erratic behaviour close to the bottom of the roughness elements due to a reduction in turbulent diffusivity [61,63,94,435]. Fig. 20 illustrates the variation of Pr_t with normal wall distance for various roughness topologies and $Pr \approx 0.7$ and 1. Seven representative roughness surfaces from Refs. [58,61,94] display behaviour at varying skewness S_{sk}, Re_τ, k_s^+ values, and types of roughness. The magnitude of these changes increases with the roughness parameter k_s^+ and appears to be dependent on the roughness topology.

For $Pr \approx 0.7$ and 1, within the hydraulically smooth regime, Pr_t behaves similarly to that of a smooth surface [61,63,94,435]. However, Fig. 20 shows that for large values of k_s^+, Pr_t exhibits a peak up to 4.0 just below the plane $y/\delta = 0$, indicating significantly reduced effective thermal diffusivity near $y/\delta = 0$. This suggests that the region $y/\delta < 0$ acts as a thermal resistance. The peak magnitude of Pr_t increases with k_s^+ and the location of the peak appears to depend strongly on the details of the roughness topology [58,61,63,94,435]. The abrupt fluctuations in Pr_t in the vicinity of roughness ($y/\delta < 0$) are associated with variations in ν_t, as the local equilibrium state of turbulence breaks down. In this non-equilibrium region, enhanced pressure diffusion acts as an energy source near the bottom of the rough surfaces, leading to increased wall-normal Reynolds stress and Reynolds shear stress [437]. Meinders *et al.* [438] also demonstrated that the recirculation zone behind a single wall-mounted cube can act as a thermal resistance. Further from the wall, Pr_t decreases with wall distance, a feature associated with the unmixedness of the scalar [232,439]. It should be noted that Pr_t over roughness is still found to be slightly larger in rough channel cases compared to smooth channel cases in the transitional rough regime [61,94].

The presence of additional dispersive stresses and fluxes in the mean transport equations, which are omitted in the classical definition (Eq. (71)), has led to the introduction of an "effective" Prandtl number [63]:

$$Pr_{t,eff} = \frac{\nu_t + \widetilde{\nu}_t}{\alpha_t + \widetilde{\alpha}_t} = \frac{\left(\langle \overline{u'_i u'_j} \rangle + \langle \widetilde{u}'_i \widetilde{u}'_j \rangle\right) \langle \overline{S}_{ij} \rangle}{2\langle \overline{S}_{kl} \rangle \langle \overline{S}_{kl} \rangle} \times \frac{\left(\left\langle \frac{\partial \overline{\theta}}{\partial x_m} \right\rangle \right)^2}{\left(\langle \overline{u'_i \theta'} \rangle + \langle \widetilde{u}'_i \widetilde{\theta}' \rangle\right) \left\langle \frac{\partial \overline{\theta}}{\partial x_i} \right\rangle} \tag{72}$$

Including dispersive stresses and fluxes is expected to make $Pr_{t,eff}$ behave more consistently, even in regions near the wall where stochastic stresses and fluxes diminish. This inclusion is also crucial for accurately modelling closure relations [424]. Kuwata [63] and Hantsis & Piomelli [58] found $Pr_{t,eff}$ and Pr_t are similar in the roughness sublayer, collapsing above the roughness crest. However, $Pr_{t,eff}$ exhibited erratic behaviour near the bottom with a value larger than Pr_t for $Pr \approx 1$. Peaks in $Pr_{t,eff}$ can be higher and occur closer to the roughness crest compared to Pr_t (for more details see [58,63,436]).

Based on all the studies [58,61,63,94], it can be concluded that, both $Pr_{t,eff}$ and Pr_t may be ineffective for modelling generalised rough-wall flows, indicating the need for alternative approaches. For instance, in smooth-wall flows, the time-scale ratio, which relates scalar variance to turbulent kinetic energy, has shown to be more useful and could potentially replace Pr_t [440]. A similar approach could be proposed for rough-wall flow cases, but limited published research on $Pr_{t,eff}$ leaves uncertainty regarding whether these observations are specific or more general, necessitating further study to confirm their broader applicability.

12. Turbulent statistics

Surface roughness has a profound impact on both turbulent velocity and temperature fluctuations, playing a crucial role in the dynamics of scalar transport. In turbulent flows over rough surfaces, the roughness elements disrupt the coherent flow structures (CFSs), such as velocity and temperature streaks, that are typically observed in smooth-wall turbulence. This disruption leads to increased turbulence isotropy and modifies the distribution and intensity of turbulent and scalar fluctuations. Various studies [4,61–63,66,92,94,170,277] have captured these effects even below the roughness crest and have shown that roughness affects both the magnitude and peak location of scalar fluctuations. In this section, we will mainly focus on the first and second-order turbulent statistics of flow and heat transfer to provide key insights into the behaviour of turbulent heat transfer over roughness. Higher-order statistics, such as turbulent budgets and temperature fluctuation budgets, are beyond the scope of this review, as they have been recently covered in another review paper by Hantsis & Piomelli [57].

Intensity of the velocity fluctuations are considered as the lowest-order indicator of the rough walls' influence on other flow parts. The most important variables, such as streamwise velocity fluctuation profiles, $\overline{uu}^+$, and turbulent kinetic energy, $\overline{TKE}^+$, have been extensively studied for different roughness heights, skewness, effective slope, and roughness types within the regime of fully developed turbulent flow. The impact of roughness topology is generally weaker for the wall-normal, $\overline{vv}^+$, and spanwise velocity, $\overline{ww}^+$, fluctuations [62,67,92–94,166,244,247,249,250]. Irrespective of the nature of the roughness, the peaks of the velocity fluctuations profiles move slightly away from the wall due to the displacement effect of the roughness on the mean flow.

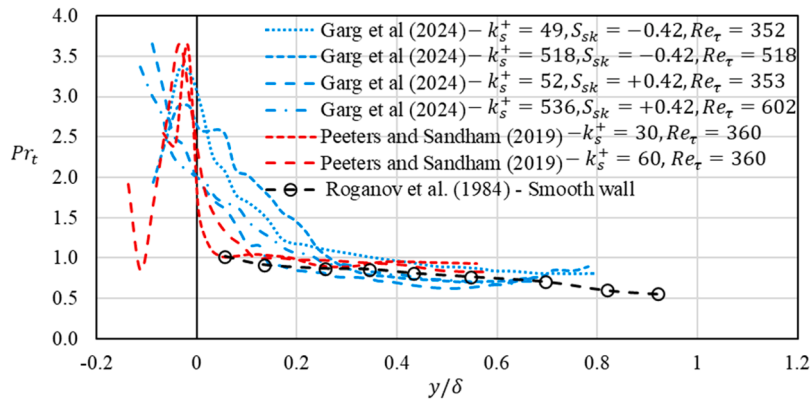

Fig. 20. Comparison of Pr_t between various roughness types and different roughness Reynolds numbers. Data was collected from Peeters and Sandham [61] (DNS, artificial irregular roughness, $Pr = 1$ for all cases), Garg et al. [94] (LES, additive-manufacture roughness, $Pr = 0.71$ for all cases), and Roganov et al. [436] (Experiment, smooth wall, $Pr = 0.71$). On the horizontal axis, the wall distance is normalized by the momentum boundary layer thickness (δ).

Moreover, the peak magnitude of $\overline{u'u'}^+$ is significantly damped (up to 50%) with increasing k_s^+ because the form drag dominates over the viscous drag, a phenomenon is widely observed in experiments [250] and simulations [62,67,92–94,166,172,441]. These studies showed that, $\overline{v'v'}^+$ and $\overline{w'w'}^+$ remain nearly unchanged, making the Reynolds-stress tensor more isotropic. Moreover, further away from the wall, the rough-wall profiles collapse onto that of the smooth wall, demonstrating that the flow is insensitive to the wall boundary and supporting the validity of the outer-layer similarity hypothesis. However, for large roughness cases, deviations from smooth wall results are observed [62,92–94], which suggests that in extreme roughness the similarity in terms of second- and higher-order statistics is not preserved, even if the similarity in the first-order turbulence statistics is clearly preserved. Similar to velocity fluctuations, numerical studies show that both the magnitude and peak location of temperature fluctuations, $\overline{\theta'\theta'}^+$, where temperature is treated as a passive scalar, are also affected by the presence of roughness. For $Pr \leq 1$, the peak value over roughs is lower than that of smooth walls and the peak location shifts outwards [61,67], but less than that of streamwise velocity fluctuations [67,442].

To better understand the relation between turbulent transfer and heat enhancement, recent studies [58,59,203,256,92,94] have focused on the comparison of time-averaged behaviour of the shear stresses, $\overline{u_i'u_j'}^+$, and heat flux, $\overline{u_i'\theta'}^+$, fields. For a wide range of k_s^+, S_{sk}, different types of roughness (regular, irregular, additively manufactured roughness), and $Pr \approx 1$, the profiles of wall-normal components of heat flux, $\overline{v'\theta'}^+$, are found to be slightly less affected by the roughness compared to the Reynolds shear stresses, $\overline{v'u'}^+$, profiles. The differences between the two are visible inside the roughness sublayer and the difference increases with roughness size, as shown in Fig. 21. Similar results in the profiles have also been observed by Garg et al. [92,94] for additive manufacture roughness and by Hantsis and Piomelli [67] for random ellipsoid roughness, although a slight difference in the magnitude is observed. However, the impact of skewness, *ES*, and other control parameters on the profiles of shear stresses and heat fluxes have received little to no attention, and further study would help in examining the viability of these results.

The main differences between the temperature and the velocity field occur in recirculation regions behind the roughness elements and in the sheltered portion of the windward slopes. This is evident in the turbulent fluxes, shown in Fig. 22, (for artificial irregular roughness [63] and additive manufacture roughness [94]), where the time-averaged wall-normal Reynolds shear stress, $\overline{v'u'}^+$, and wall-normal heat flux, $\overline{v'\theta'}^+$, are shown in a vertical plane (a-b) for $k_s^+ \approx 200$ and $Pr = 1$; and (c-d) for $k_s^+ \approx 500$ and $Pr = 0.7$. For the most part, the velocity and temperature fields appear visually similar, which is anticipated since both are driven by the same turbulent eddies and are associated with the wall-normal velocity component, v'. However, there are noticeable differences in the recirculation regions downstream of larger roughness elements, where the heat flux is more intense than the shear stress. It is speculated that the strengthening of both turbulent stress and flux close to the rough surface is due to the development of quasi-streamwise elongated vortices in the narrow gaps within the roughness elements [63,92,94].

Leonardi et al. [64], Doosttalab et al. [435], and Peters and Sandham [61] have attempted to provide an additional method to examine the relationship between the flow topology and the heat fluxes by computing the probabilities of turbulent events, as well as their relative contributions to the Reynolds stresses and heat fluxes. Wallace et al. [443] employed a quadrant analysis, motivated by the desire to gain insight into the near-wall streaks, such as events where the streaks slowed down and then erupted away from the wall, and the instances where large-scale structures far from the wall move towards it. This method can be used to examine the instantaneous behaviour through the probabilities of turbulent events, as well as their relative contribution to the Reynold stresses and scalar fluxes.

The quadrants are obtained by utilising the signs for the streamwise and wall-normal velocity fluctuations and are defined as follows: q_1: $u' > 0$, $v' > 0$; q_2: $u' < 0$, $v' > 0$; q_3: $u' < 0$, $v' < 0$; and q_4: $u' > 0$, $v' < 0$. In flat-plate boundary layers and plane channels, q_2 events represent fluid moving away from the wall towards the outer flow, while q_4 events represent outer-layer fluid being advected towards the wall. They are the main contributors to the Reynolds shear stresses. For a detailed review on quadrant analysis and its field of applicability, the reader is referred to Wallace et al. [443]. Replacing u' with θ' in q_1 to q_4 provides an additional method to examine the relationship between flow topology and heat fluxes. In rough surface flow studies, u' and v' are considered to be fluctuations parallel and normal to the wall at the bottom of the roughness layer, rather than to the roughness itself. Consequently, q_2 events represent fluid moving away from the roughness layer towards the outer flow, while q_4 events represent outer-layer fluid being advected into the valleys of the roughness.

Fig. 23 illustrates the quadrant analysis, in the near wall region, adopted from Peeters and Sandham [61]. Above the roughness crest, the smooth-wall distribution is found both in the $u' - v'$ and $v' - \theta'$ planes, with q_2 and q_4 events dominating, as shown Fig. 22 (c) and (d). On the other hand, inside the roughness sublayer, the probability of the four quadrants becomes roughly equal for the stress, tending to be roughly 25%, as the base of the roughness is approached. For the heat flux, however, the probability of q_4 events, where $\theta' > 0$, and $v' < 0$ is meaningfully diminished, while the probability of q_3 events, where $\theta' < 0$ and $v' > 0$, is significantly enhanced [61,64], as shown in Fig. 22

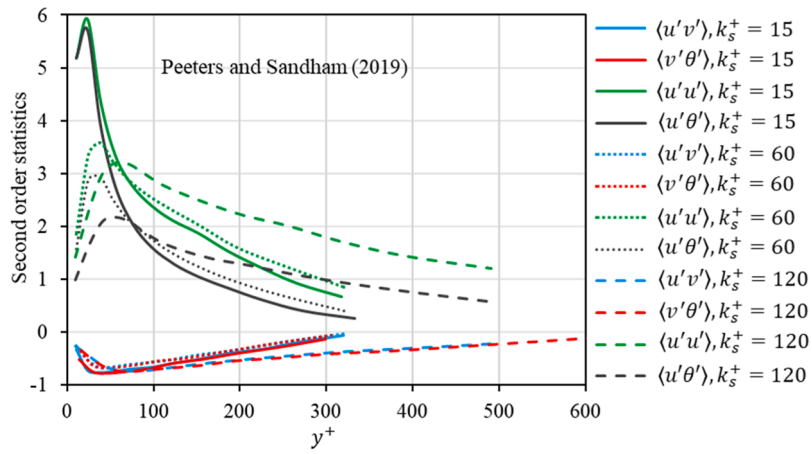

Fig. 21. Comparison of wall-normal Reynolds stress, $\overline{v'u'}$, streamwise fluctuations, $\overline{u'u'}$, normalized by u_τ^2 and heat flux, $\overline{u'\theta'}$ and $\overline{v'\theta'}$, normalized by $u_\tau T_\tau$ for cases with different values of k_s^+. The data is adapted from Peeters and Sandham [61] for $Pr = 1$.

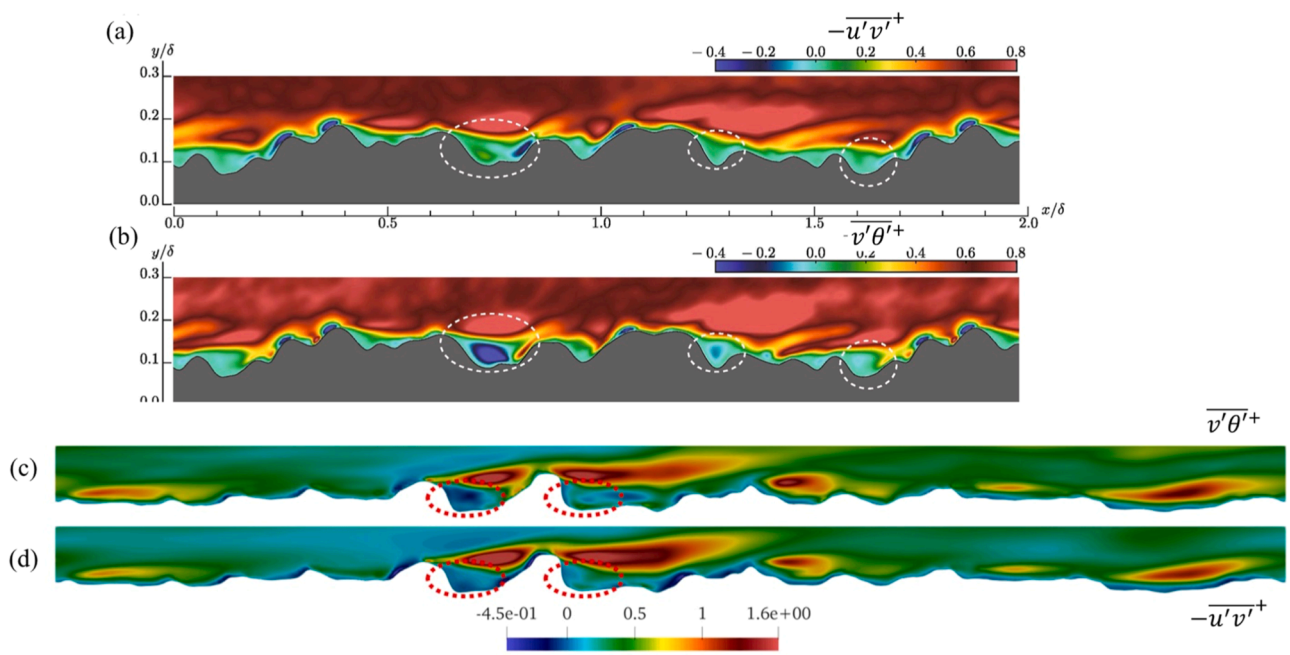

Fig. 22. Contour maps of time-averaged wall-normal heat flux components and Reynolds shear stress in a random *x*-*y* plane. (a)-(b) artificial irregular roughness adapted from Kuwata [63] by the permission of Elsevier; and (c)-(d) additively manufactured roughness adapted from Garg *et al.* [94].

(d). The difference in event probabilities, however, does not translate into the contribution to the mean flux and stress [58,444]. Examining the relative contributions to mean heat flux, $\overline{(v'\theta')_j}/\overline{(v'\theta')}$, shows q_4 events contribute more than q_2 events. This means that fewer q_4 events occur but they are more intense compared to the smooth-wall case [54, 58,428], showcasing that roughness enhances the transport of both stochastic Reynolds stresses and heat fluxes towards the valleys and increases u' and θ' skewness. Despite the conclusion that the q_1 to q_4 events contributions to mean values are similar for momentum and scalar fluxes, there are notable differences in the types of motions each experience, indicating a local breakdown in the analogy between momentum and scalar transport in the near-wall region [61]. More details on the quadrant analysis can be found in Peeters and Sandham [61].

13. Conclusion

Despite substantial advancements in understanding friction drag, heat transfer over rough walls has received comparatively less attention. The majority of studies have concentrated on the fully rough regime, while investigations into heat transfer in transitionally rough and hydraulically smooth regimes remain sparse. The study of flow over rough surfaces presents a complex problem characterised by a vast parameter space, surpassing that of momentum transfer. Both roughness topology and the Prandtl number, *Pr,* significantly influence heat transfer, with the *Pr* exerting a particularly pronounced effect. However, existing research is limited to a narrow range of roughness types and *Pr*. This review summarises the progress made in rough-wall heat transfer, highlights the current limitations, and identifies critical gaps in the literature, setting the stage for future investigations.

A substantial portion of the literature examines how roughness topology affects convective heat transfer. Most research has focused on the impact of roughness height, while fewer studies have explored the effects of spatial parameters, such as effective slope *ES*, and frontal solidity, λ_f, on heat transfer characteristics. Among other roughness parameters, roughness density has a significant influence on heat transfer. Roughness density affects the impingement of flow on the

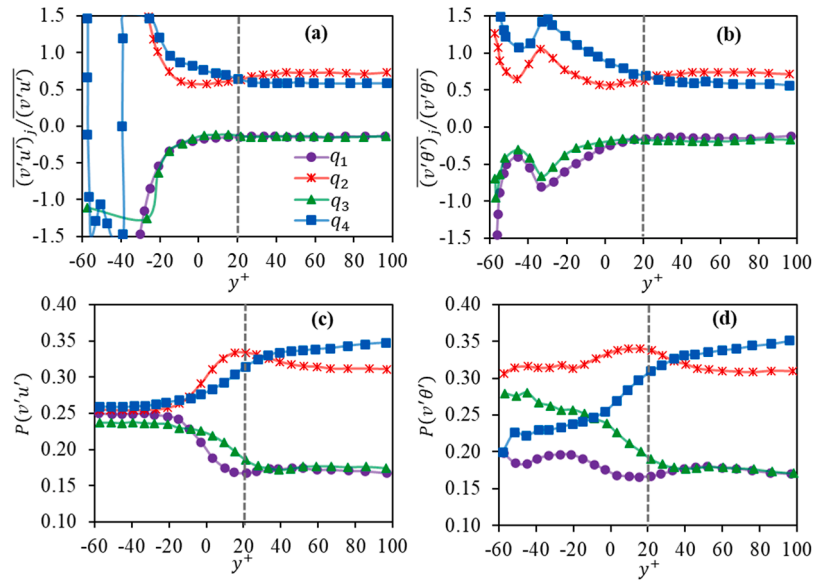

Fig. 23. Quadrant analysis for the Reynolds stress, $\overline{v'u'}$, and scalar flux, $\overline{v'\theta'}$, at $k_s^+ = 60$. (a,b) Relative distributions to the Reynolds stress and scalar flux, respectively and (c,d) probabilities of the various events. The vertical dashed line indicates $k_S/\delta = 1/3$. The data is adopted from Peeters and Sandham [61].

windward surfaces of roughness elements, a crucial mechanism driving heat transfer enhancement. Within the transitional roughness regime, for $0.1 < \lambda_f < 0.3$, heat transfer tends to decrease with increasing λ_f. Conversely, in the fully rough regime, the trend reverses, and heat transfer increases with λ_f. When $\lambda_f \geq 0.3$, sheltering effects dominate, reducing both friction and heat transfer.

Research in rough-wall heat transfer is primarily based on two theoretical models: the "cavity-vortex hypothesis" and "Kolmogorov-Brutsaert theory". The cavity-vortex hypothesis has been widely adopted in heat transfer applications, whereas surface renewal theory is more commonly used in the context of scalar transport, such as in meteorology and ABL flows. These theories have been leveraged by various studies to develop engineering models and correlation for thermal boundary layer. Despite these limitations, these theories have informed various engineering models and correlations for the thermal boundary layer. Current literature suggests that St_k scales as $(\lambda_f)^n (k_s^+)^{-p} Pr^{-m}$, but there is ongoing debate about the appropriate values of the exponents. While many studies recommend $n = p = 1/2$ and $m = 2/3$ significant discrepancies remain between the empirical correlations derived from these models. These inconsistencies often arise from empirical fitting within a restricted parameter space (k_s^+, Pr), insufficient consideration of roughness topology, and the absence of a unified scaling approach.

Several studies have explored the temperature shift, $\Delta\Theta^+$, as a measure of heat transfer enhancement due to roughness, similar to how the roughness function ΔU^+ is used for momentum transfer. While the variation of ΔU^+ with k_s^+ shows an asymptotic behaviour within the fully rough regime, with data for different flow and roughness types collapsing onto a single curve, such behaviour has not been observed for $\Delta\Theta^+$. Current data show an increase in $\Delta\Theta^+$ with k_s^+ in the transitionally rough regime, but the behaviour of $\Delta\Theta^+$ in the fully rough regime remains poorly understood. Some studies suggested that $\Delta\Theta^+$ may exhibit an asymptotic behaviour with increasing k_s^+ in the fully rough regime. Conversely, there is evidence indicating a reduction in $\Delta\Theta^+$ with k_s^+ after it peaks at a specific value in the fully rough regime. If this decreasing trend persists, it could imply that $\Delta\Theta^+$ might eventually become negative, suggesting a reduction in heat transfer compared to a smooth wall. The current heat transfer models do not account for either trend, indicating a need for a new theory or a revision of the traditional models.

The mechanisms governing heat transfer over rough surfaces are not well understood. From the perspective of turbulence theory, roughness-scale eddies increase with roughness height, leading to greater interactions among turbulent scales and enhanced heat transfer. Phenomenologically, roughness that extends into the higher-velocity flow region facilitates the entrainment of fresh fluid from the bulk flow into the roughness layer, thereby enhancing heat transfer. This suggests the importance of roughness thermal conductivity and its impact of energy transfer from solid wall to the fluid. This mechanism has been hypothesized as a significant contributor to heat transfer in rough-wall laminar flows [91,445].

Heat transfer enhancement by roughness is always accompanied by increased flow friction. Most of the literature reports that the Reynold analogy efficiency, RA_R/RA_S, is less than 1, suggesting that heat transfer enhancement never exceeds the corresponding increase in friction. However, these findings are limited to certain roughness topologies and flows with Prandtl numbers of 0.7 or 1.0. It can be hypothesized that, in the transitionally rough regime, with a sufficiently high Prandtl number, RA_R/RA_S greater than 1.0 could be achievable.

Higher-order statistics of rough-wall heat transfer have received limited attention. Literature findings demonstrate that both velocity and temperature fluctuations, when scaled in inner units, are attenuated by roughness, with velocity fluctuations being more significantly affected than temperature fluctuations. This difference is likely due to variations in the production mechanisms of velocity and temperature fluctuations. It was also found that the mean vertical turbulent heat flux, $\overline{v'\theta'}^+$, is larger than the mean shear stress, $\overline{v'u'}^+$, for $Pr \approx 1$. This suggests that vertical heat flux plays a significant role inside the roughness sublayer and is important for the production of turbulent fluxes. Furthermore, the turbulent Prandtl number, Pr_t, behaves unpredictably near the rough surface, indicating that it may not be a reliable tool for modelling generalised rough-wall flows.

13.1. Limitations

Some limitations in numerical studies arise from the application of surface boundary conditions, as well as the restricted range of Reynolds and Prandtl numbers considered. The uniform surface temperature boundary condition is the most commonly used due to its convenient implementation with periodic boundary conditions. The mixed-boundary condition has also been employed in some studies to simulate the isoflux condition, although this assumes that the wall remains

locally isothermal. However, these boundary conditions are limited to a narrow range of Prandtl numbers and do not account for temperature variation on the surface. In reality, the surface temperature fluctuates, with peaks and valleys that vary over time.

Scale-resolving approaches are computationally expensive. The computational cost for resolving the momentum field in smooth-wall DNS and LES flows is typically on the order of $O(Re_\tau^{9/4})$ [189,446]. For simulations involving $Pr > 1.0$, the cost increases to $O(Re_\tau^{9/4} Pr^{3/2})$ [447]. When roughness is introduced, the required computational grid to accurately resolve the roughness can exceed that required for true DNS. Due to these computational cost limitations, only a limited number of studies have achieved $k_s^+ \approx 500$, while many remain within the transitionally rough regime.

Another limitation lies in the extensive variety of roughness geometries. Roughness geometry is arguably the most challenging parameter space to cover. The literature spans a wide range of roughness types (such as regular, irregular, 2D, and 3D), as well as various morphological characteristics within each type. Most studies tend to focus on a single roughness type, adjusting primarily the roughness Reynolds number (often by increasing the Reynolds number) and typically considering just one value of the Prandtl number, usually around 1.0.

There is a notable lack of experimental heat transfer studies in the literature. This gap is likely due to the large uncertainties associated with both measurement and processing techniques, as highlighted by Rowin et al. [129]. Experimental measurements of convective heat transfer are influenced by various factors, such as entrance effects, roughness effects, and instrumentation uncertainties and errors [448]. Even highly accurate measurements can be significantly affected by uncertainties in geometry measurements [449]. To improve the accuracy of measurements on heated rough walls, several recommendations should be followed, including directly measuring wall shear stress and heat flux, selecting heated plates with low emissivity, and ensuring precise control over the heated wall conditions [129].

13.2. Future studies

The literature lacks a systematic study on how heat transfer scale with various roughness topology. To determine whether any observed findings or trends exhibit a degree of universality, a broader and more systematic exploration of this parameter space is essential. Further studies are needed to cover a wider range of (k_s^+, Pr) systematically varying roughness parameters across different roughness types. Accurate characterization of these quantities is crucial for numerical simulations, as they present challenges for experimental measurement.

Numerical studies have primarily focused on turbulent flow characteristics, but several critical aspects, such as the impact of boundary conditions, wall origin, and turbulent statistics, remain largely underexplored. The influence of surface boundary conditions has received limited attention in the literature, mainly due to the challenges in defining wall boundary conditions for periodic domains, as discussed in Section 13.1. New numerical approaches need to be modified to address these limitations. Additionally, most studies assume the same origin for both heat and momentum transfer, which is not always valid, and this assumption warrants further investigation.

There is a pressing need for conjugate heat transfer studies to investigate the mechanisms of energy transfer from the solid wall to the fluid and to understand the impact of roughness thermal conductivity.

Experimental studies, particularly those involving irregular roughness, are scarce. Further experimental and numerical studies are required to scale heat transfer characteristics with rough flow parameters, especially for $k_s^+ > 500$, and $Pr \gg 1$ or $Pr \ll 1$.

Despite the substantial research gap in understanding average flow and heat transfer over rough surfaces, there exists an even wider gap in the turbulent flow statistics, particularly the behaviour of the turbulent Prandtl number and second- and higher-order turbulent statistics.

CRediT authorship contribution statement

Mohammadreza Kadivar: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Himani Garg:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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